

A COMPETITIVE BOOK OF

Agricultural Entomology

Seventh Edition

**A book for all your Agricultural
Competitive exams**

Published by:

Agri Coaching Chandigarh

Index

S. No.	Topic	Page No.
1	Basics of Entomology	68
2	Classification of Insects	69
3	Orders of Insects	75
4	Different kind of Organs	81
5	Modifications	85
6	Terminology in Entomology	87
7	History	88
8	Apiculture	89
9	Terms related with apiculture	93
10	Sericulture	94
11	Lac Culture	94
12	Some Important Polyphagous pests	97
13	Methods of Pest Control	102
14	Different classifications of Insecticides	107
15	Pests of field crops	111
16	Insect -Pests of Pulses	121
17	Diseases and Enemies of Honeybees	123
18	Diseases of Silkworm	126
19	Insect Pests of Fruit Crops	129
20	Insect-pests of Temperate Fruits	138
21	Some Important Insects of Major Crops	143
22	Facts from Previous Year Exams	146

ENTOMOLOGY:

- Entomology is a branch of zoology, which deals with the study of insects.
- **Entomon** = Insects, **Logos** = study.

AGRICULTURAL ENTOMOLOGY:

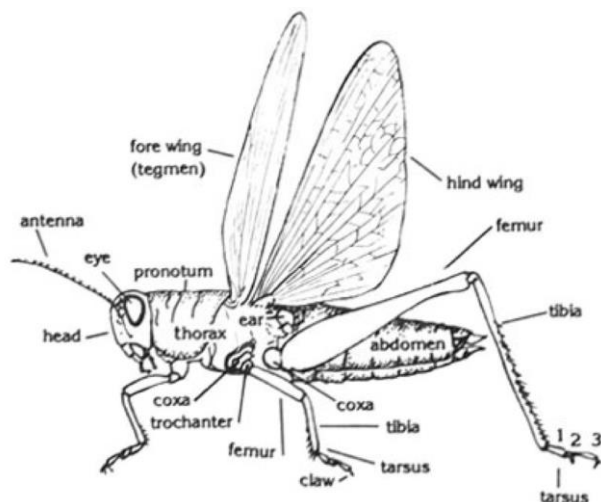
- This mainly concerns with the study of insects, which are directly related with the crops and the stored commodities.

❖ Arthropoda

- **Arthros** = means jointed, **Poda** = legs/ appendages.
- They are characterized as having a body divided by grooves to form segments and a well-developed covering, the **Integument**, which makes up the outer shell or **Exoskeleton**.

❖ INSECT:

- Insect is an arthropod having body segmented into three regions- head, body and thorax and have 3 pairs of legs and 2 pairs of wings or Insects are **invertebrates** having body segmented into three regions. They have bilateral symmetry and grow by moulting.



- Insects may live in water, on land or in soil, during part of or all of their lives.
- Their lifestyle may be **solitary, gregarious or highly social**.

❖ PEST:

- Pests are the organism belonging to class insecta causing economic loss in plants.

- Any organism which causes economic damage to the crops or interferes with the welfare of human beings or his belongings.
- **Pest** is derived from term “**pestis**” which means **plague**.
- Generally, any organism which causes irritation to the other organism.

❖ EXOSKELETON

- The exoskeleton not only provides a larger area for the attachment of muscles but also protects muscles from mechanical injury.
- It affords excellent mechanism to protect insects against desiccation.
- In some insects, the exoskeleton has turned appendages into good tools for digging, preying, oviposition etc.
- The exoskeleton maintains the shape of the body much more efficiently than endoskeleton.



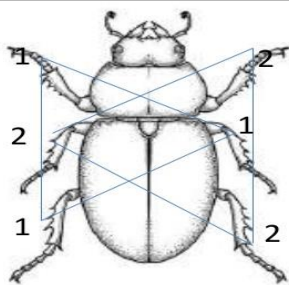
❖ FUNCTIONAL WINGS

- It has increased the feeding and breeding range and provided a new means of eluding enemies of attacking a fast-moving host and of finding a mate.
- Increasing feeding range undoubtedly opened the way for the adoption of specific food especially in those areas in which the host or breeding medium occurred in small quantities and in scattered situations.
- E.g., Mosquitoes, which without wings would not have been able to breed in small pools of water which dry in a short time.
- Intercontinental migrations in locusts and butterfly are very common.

❖ HEXAPOD LOCOMOTION:

- Jointed nature of legs is a characteristic which insects share with their other Arthropod allies, but the number of such legs has undergone an ideal stage of evolution in insects.
- Six legs in insects represent the optimum number which no other group of animals has hit upon.
- The large no. of legs as in millipedes is bound to make the locomotion rather cumbersome but when the number of walking legs become less than six, it is bound to create a problem of balancing during locomotion.
- Thus, six is the smallest number for a stable equilibrium during all stages of terrestrial locomotion.

10. Hexapod locomotion: Insects use 3 legs at a time during locomotion, while the remaining 3 legs are static, which gives greater stability.



❖ COMPOUND EYES

- Insects usually have a pair of compound eyes in nymphal and adult stages and in addition sometimes simple eyes too.
- Their eyes are called compound because each eye instead of having a single large cornea, consists of a number of hexagonal areas each representing the cornea of a discrete visual organ called the ommatidium all of which are compacted together.
- E.g., Ants have 50-400 facets or corneas in each eye, the housefly -4000 dragonfly > 50,000. It is obvious that the insect will not lose this power of vision completely if a few of their ommatidia are injured.

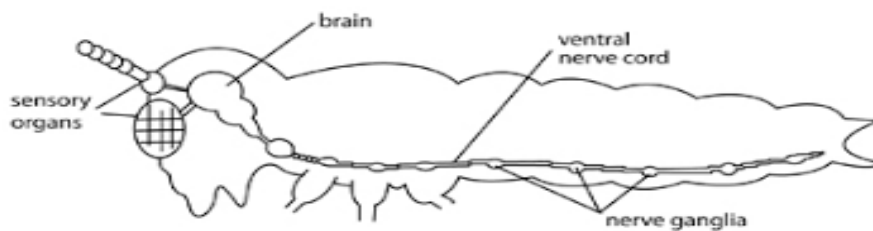
❖ Scattered sense organs

- Except the eyes, none of the other sense organs are invariably concentrated on the head. The organs of hearing are sometimes located on the antennae.
- Similarly, the organs of taste and smell may occur on the antennae, mouth parts, or even on the tarsi or cerci. Such diffused and scattered nature of sense organs are bound to be

advantages to insects as the arrangement reduces the chances of all receiving injury at the same time.

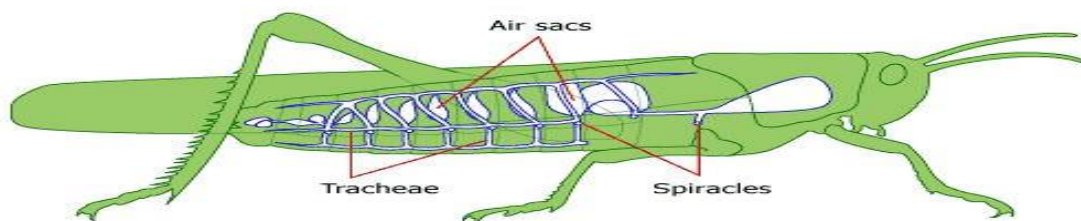
❖ Decentralized nervous system

- The central nervous system is a ladder like **chain of ganglia** along the ventral side of the body except one is also dorsal to alimentary canal.
- This system is so decentralized that brainless insects can be artificially stimulated to walk, fly or even feed and according to some, the living abdomen of the female silkworm moth when separated from the thorax can be fertilized by the male and stimulated to lay fertile eggs.



❖ Direct respiration

- This important physiological function is independent of the mediation of blood which in insects is not required to transmit the gases of respiration.
- **These gases are taken to and fro every minute part of the system directly through air tubes called trachea and tracheoles which have several openings called spiracle rather than a pair of nostrils.**



❖ Enteronephric excretion

- The excretory function is performed by **Malpighian tubules**, which open into the front part of hind intestine instead of opening directly to the exterior. Also, the excretory products are semisolid rather than liquid urine.
- **Predators** are **free-living** species that directly consume a large number of **preys** during its whole lifetime.
- Ladybird beetles, specifically their **larvae are active** predators on aphids, and mites, scale insects and small caterpillars.
- **Dragonflies** are important predators of **mosquitoes**.
- Parasitoids lay their eggs on or in the body of the insect host, which is then used as food for the developing larvae.
- Most parasitoids are wasps and flies, and usually have a very narrow host range.
- Each insect species has essential nutritional requirements for the completion of its life cycle.

❖ **Based on nutritional requirements, the insects are classified as below**

- 1. Carnivorous:** They feed and live on other animals as parasites and predators.
- 2. Omnivorous:** They feed on both plants and animal such as wasps.
- 3. Herbivorous:** They use living plants as their food such as various crop plants.

❖ **The insect of this category is further classified as-**

- **Monophagous:** The insect which feed on a single species of plant. E.g., **Brinjal fruit borer, Pink boll worm in cotton**.
- **Oligophagous:** In this case the feeding activities of a particular species of insects are confined only to a plant of one botanical family. **E.g., Cabbage butterfly, mustard saw fly**.
- **Polyphagous:** The insect which feed on various types of cultivated and wild plants. **E.g., Bihar hairy caterpillar, Gram pod borer, Locust, Grasshopper etc.**

❖ **Economic importance of insects**

- All insects are not harmful, those harmful are not harmful unless their number cross certain limit.
- Economic importance of insects lies more in their harmful effects than beneficial effects.

- Higher the status of an insect as pest, more important that insect is.

❖ **Beneficial effects:**

a. Insects of industrial importance:

- **Honeybees:** Honey, beeswax, bee venom, royal jelly, propolis and Pollination
- **Silkworm:** Silk
- **Lac insect:** Shellac

b. Pollination: Transfer of pollens from anthers to stigma

- **Insect Pollinators:** Honeybees (75-80%), butterflies, flies, beetles, thrips, **etc.**

c. Entomophagous insects:

(i) Predators:

- **Coccinellids (Coleoptera): e.g.,** *Coccinella septempunctata*, *Hippodamia variegata*, *Menochilus sexmacula*, etc.
- **Syrphids (Diptera): e.g.,** *Episyrphus balteatus*, *Metasyrphus corollae*, *Scaeva pyrastris*
- **Chrysopids (Neuroptera): e.g.,** *Chrysoperla carnea*
- **Predatory bugs (Heteroptera): e.g.,** *Anthocoris minki*, *Orius spp*
- **Preying mantids (Dictyoptera): e.g.,** *Statilia maculata*
- **Predatory wasps (Hymenoptera): e.g.,** *Vespa spp*

(ii) Parasitoids:

- **Egg parasitoid: e.g.** *Trichogramma spp.*
- **Larval parasitoid: e.g.,** *Cotesia (Apanteles) spp.*, *Bracon sp.*
- **Egg-larval parasitoid: e.g.,** *Chelonus blackburni*,
- **Larval-pupal parasitoid: e.g.** *Ceromisia auricaudata* (**Diptera: Tachinidae**)
- **Pupal parasitoid:** *Brachymeria nephontidis* on *Earias spp*
- **Adult parasitoid:** *Blaesoziphae kellyi*

- **Egg parasitoid:** attacks egg stage of the host. Example- **Trichogramma chilonus**, **T. preteosum**, **Ooencyrtus** sp. a true egg parasitoid of various **lepidopterous** pests.
- **Egg Larval Parasitoid:** Another category of egg parasitoids, which deposits eggs in the host eggs, but development and emergence are not completed until the host larval stage is reached. e.g., **Braconid**, **Chelonus annulipes** on **European corn borer**, **Chelonus blackburni** on **PTM** and **bollworm**, **Capidosoma koehleri** on **PTM**.
- **Larval Parasitoids:** True larval parasitoids are those parasitoids which deposit egg on the larvae and their progeny complete development and emerge from the host larvae. e.g., **Apanteles spp.** On **pink bollworm**, **Bracon brevicornis** on **spotted boll worms**.
- **Pupal Parasitoid:** Parasitoids that deposit their eggs in the host pupae and emerge from the host pupae. e.g., *Brachymeria nephontidis* on *Earias* spp. All *Chalidae* (Hymenoptera) are pupal parasitoids.
- **Adult Parasitoids:** Parasitoids of adults hosts. The parasitoids deposit a larva on its host while in flight and the mature maggot emerges from the dead adults host.' e.g., *Blaesoziphae kellyi* a parasitoid of locust.
- **Nymphal Adult Parasitoids:** The parasitoids deposit their eggs on/in host nymph and emerge from the dead adult host. e.g., *Epiricania melanoleuca* parasitic on *Pyrilla perpusilla*.
- **Nymphal Parasitoids:** The parasitoids deposit eggs on nymphs and their progeny complete development and emerge from the host numphs. e.g., *Epipyrops fuliginosa* on nymphs of *Idioscopus clyealis*.

d. Nutrient cycling:

- **Detritus** and dung feeders e.g., Termites, dung beetles etc.

e. Human food:

- **Over** 500 spp. like crickets, grasshoppers, beetles, caterpillars, termites are used as food by human beings in different parts of the world.

f. Aesthetic value:

- **Brightly** coloured butterflies and beetles are used for various decorations

❖ Harmful effects:

- Only a few species (0.01%) are harmful and attain the status of pest.

- Higher the status of species as pest more important the species is.
- Crop losses by the insect pests vary from 10-40 %

❖ **Different pesticides used:**

Insecticides	60%
Fungicides	19%
Herbicides	16%
Bio pesticides	2%
Others	3%

❖ **ORDERS OF INSECTS:**

Chilopoda	(Chilo - lip; poda - appendage) e.g. Centipedes
Diplopoda	(Diplo - two; poda - appendage) e.g. Millipede
Crustacea	(Crusta - shell) e.g. Prawn, crab, wood louse
Arachnida	(Arachne - spider) e.g. Scorpion, spider, tick, mite

❖ **CHARACTERS OF DIFFERENT ORDERS:**

Characters	Crustacea	Arachnida
Habit	Aquatic and few terrestrial	Terrestrial
Antenna	2 pair	No antenna
Visual organs	One pair Compound eyes	One pair-simple eyes
Locomotor organs	Minimum five pairs of legs	Four pairs
Respiration	Gill breathing	Book lungs (Scorpion) and tracheal(spiders)
Body fluid	Hemolymph	Hemolymph
Circulatory system	Heart with ostia	Heart with ostia

Characters	Chilopoda	Diplopoda	Hexapoda
Habit	Terrestrial	Terrestrial	Many terrestrial

			and very few aquatic
Antenna	One pair	One pair	One pair
Visual organs	One pair-simple eyes	One pair-simple eyes	Both simple eyes and compound eyes (one pair)
Locomotor organs	One pair per segment (First pair of legs modified as poison claws)	Two pair per segment (No poison claws)	
Respiration	Tracheal	Tracheal	Tracheal
Body fluid	Hemolymph	Hemolymph	Hemolymph
Circulatory system	Heart with ostia	Heart with ostia	Heart with ostia

❖ **Class Insecta has 2 sub-classes:**

1. Apterygota
2. Pterygota

❖ **Pterygota is further divided into**

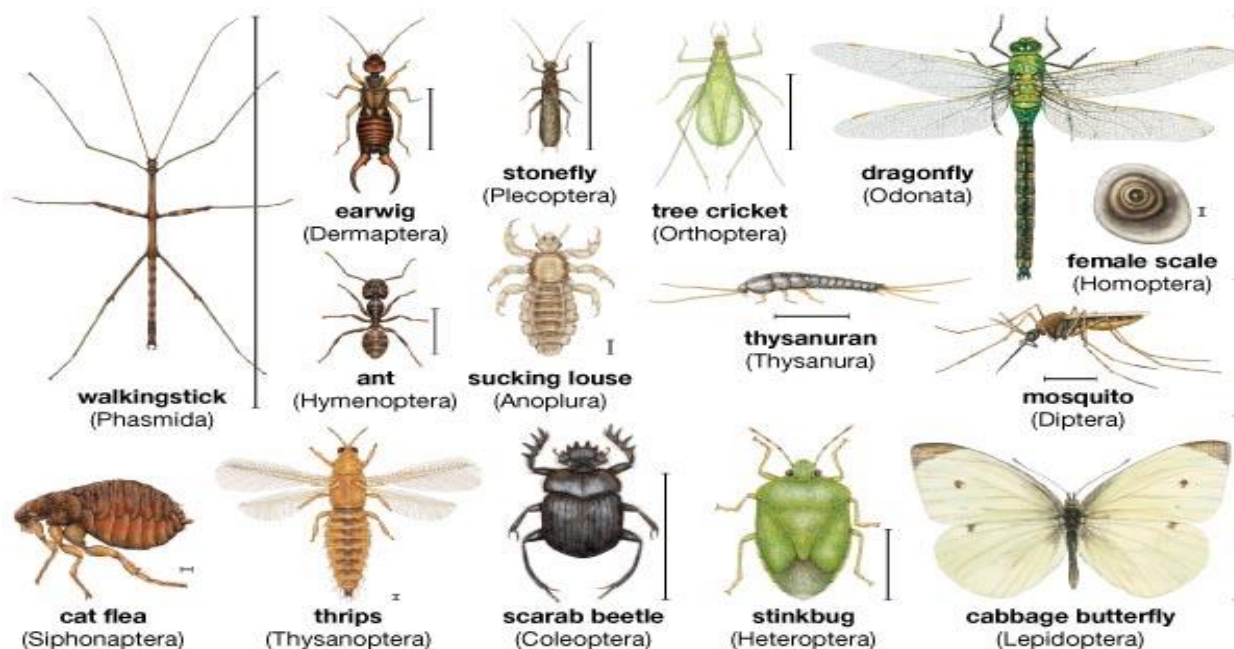
- Exopterygota
- Endopterygota

❖ **Apterygota:**

- Primarily wingless, metamorphosis absent, pregenital appendages present.
- It includes the orders:
 - ✓ **Protura (telsontails)**
 - ✓ **Diplura (japygids)**
 - ✓ **Collembola (springtails)**
 - ✓ **Thysanura (silverfish)**

❖ **Exopterygota-**

- Wings developed externally, metamorphosis incomplete, pupa absent, immature nymphal stage present.
- Incomplete metamorphosis- presence of 3 stages; eggs, nymph and adult.



❖ Endopterygota:

- Wings developed internally, metamorphosis complete, pupa present, immature stage is larva.
- **Complete metamorphosis**- presence of 4 stages; egg, larva, pupa and adult.

❖ Orders of Exopterygota with their examples:

Orders	Examples
Orthoptera	Locust, Grasshopper
Isoptera	Termite
Thysanoptera	Thrips
Hemiptera	1. Heteroptera: True Bugs 2. Homoptera: Aphid, Leaf Hopper

❖ Orders of Endopterygota with their examples:

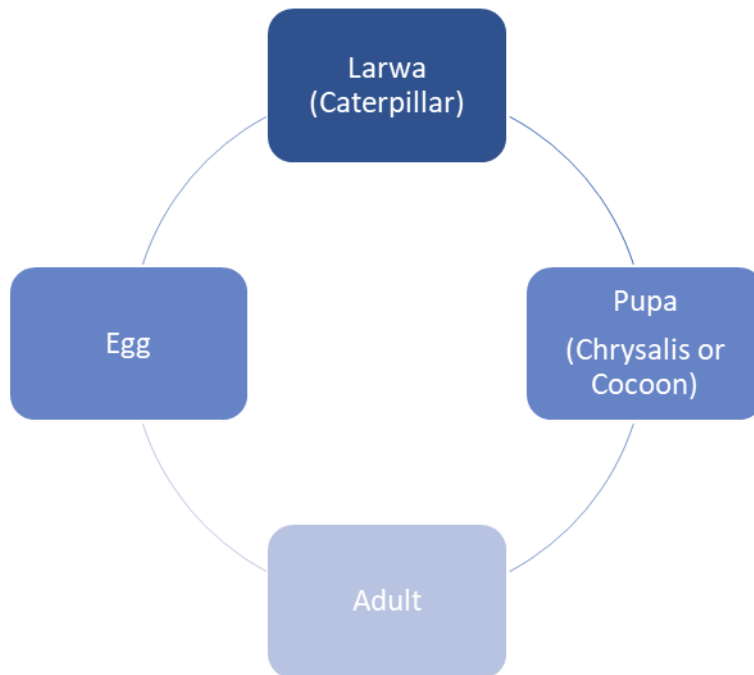
Orders	Examples
Coleoptera	Beetles and Weevils (This order is most damaging)
Lepidoptera	Butterflies and Moths
Diptera	Flies
Hymenoptera	Honeybee, Saw Flies, Ants, Wasps

❖ **Metamorphosis:** A profound change in form one stage to the next in the life history of an organism.

❖ **Types of metamorphosis:**

Ametabolous	No metamorphosis
Hemimetabolous	Incomplete metamorphosis
Holometabolous	Complete metamorphosis

- **Simple Metamorphosis/ Ametamorphosis:** Hatches from egg looking just like the adult grows in size by molting.
- **Incomplete Metamorphosis/ Hemi-metamorphosis:** Hatches from egg and becomes a nymph. Nymph does not have fully developed wings. Molts to become an adult.
- **Complete Metamorphosis/ Holo-metamorphosis:**



❖ **Types of Larvae:**

- **Nymph-** Larva of Orthoptera (grasshoppers/locusts) and Hemiptera (**bugs, white flies, aphids and jassids**).

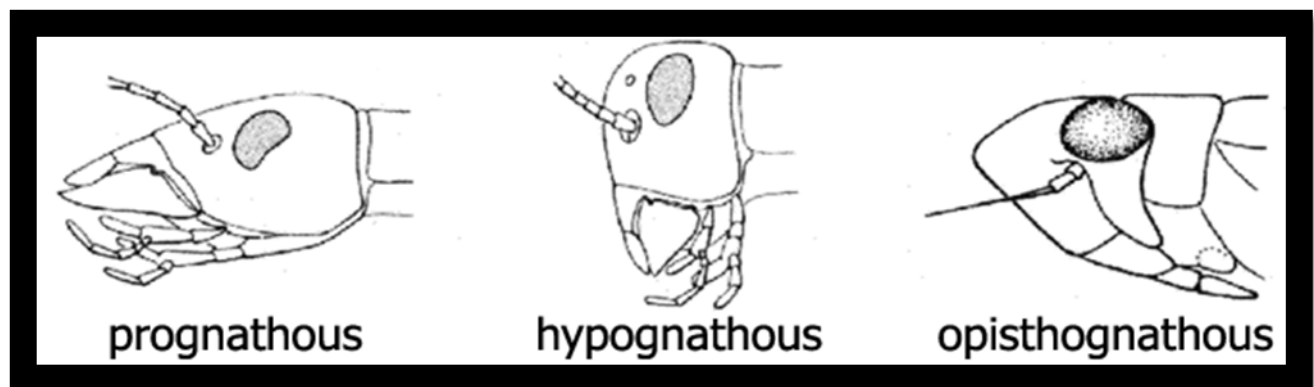
- **Caterpillar**- Larva of Lepidoptera (**butterflies/moths**), **Bollworm, borer** (except lesser grain borer).
- **Grub**- Larva of Coleoptera (**beetles/ weevils**) and Hymenoptera (**bees. Wasps, ants**).
- **Maggot**- Larva of Diptera (**true flies**)
- **Niads**- Larva of Odonata (**dragonfly/damselfly**)

❖ **The Insect Head:**

- The head of an insect is an almost completely sclerotized capsule formed by the fusion of several sclerites often distinctly seen in the **embryo**.

❖ **Based on the inclination of the long axis and the mouth parts, it is usual to distinguish three types of heads.**

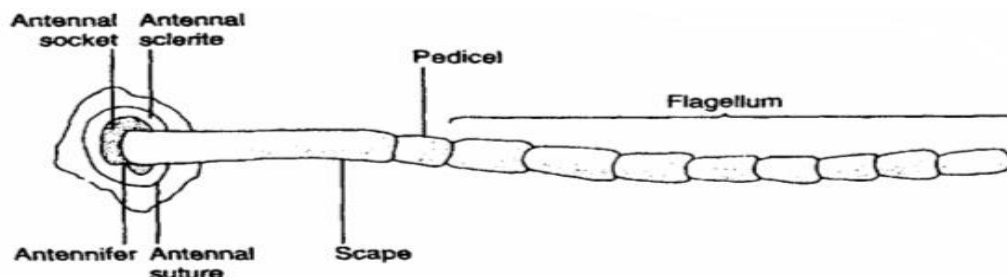
- **Hypognathous** head with the mouth parts ventrally placed the long axis of the head being vertical. The median line of the head forms right angle with the median line of the body. **Eg. Grasshopper.**
- **Prognathous head**- Here the long axis is horizontal, and mouthparts placed anteriorly. **Eg. Beetles.**
- **Opisthognathous head**- mouthparts are posteroventral (placed between the anterior pair of legs) and head directed backwards. **E.g., Bugs.**



- ❖ **Antennae**: Antennae are mobile, segmented, paired appendages. Primitively, they appear to be eight segmented in nymphs and adults, but often there are numerous subdivisions, sometimes called antennomeres.

❖ Antennae having three segments:

1. Scape
2. Pedicel
3. Flagellum



- **Scape:** it is the first segment, larger than the other segments and is the basal stalk.
- **Pedicel:** It is the second segment, nearly always contains a sensory organ known as Johnston's organ, which responds to movement of the distal part of the antenna related to the pedicel.
- **Flagellum:** It is the filamentous and multisegmented (with many flagellomeres) but may be reduced or variously modified.

❖ TYPES OF ANTENNAE:

Type	Remarks	Example
Aristate	Pouch-like	House fly
Setaceous	Bristle-like	Dragonfly/damselfly
Filiform	Thread-like	Grasshopper
Moniliform	Bead-like	Termites
Serrate	Saw-toothed	Beetles
Pectinate	Comb-like	Moths, sawflies, beetles
Capitate	Head like (less enlarged at the tip would be clavate -clublike)	Butterflies
Geniculate	Elbowed	Honeybees
Lamellate	Plate-like	Beetles
Plumose	Plumed or feather-like	Mosquito

❖ DIFFERENT KIND OF ORGANS:

1. **Organs of smell:** In some insects the smell organs (sensoria) are situated in the antennae by which they recognize their food etc. **e.g., Ants, honeybee.**

2. Organs of taste: Some insects bear taste hair on their antennae by which they know the taste of their food e.g., cockroach.

3. Stridulatorial organs: Sound producing organs are in the antennae of some insects belonging to the orders-coleopteran and orthopteran e.g., cricket.

4. Organs of chordotonal: The hearing organs often known as Johnston's organs are situated in the second segment (pedicel) of antenna e.g., male mosquito.

5. Sexual characters: In some of the insects belonging to the order Diptera and Hemiptera, the antennae are found of different type in male and female viz. mosquito.

6. Other functions: In some larvae the antennae are adapted for seizing the prey e.g., *Chaborus*, while in other insects they are used for holding the females. e.g., the male of *meloe*. The butterflies are having some transmitting and receiving organs in their antennae.

❖ Mouth parts:

- **Important** for insect identification.
- Provides information on feeding habits and types of damage

Mouth Part	Remarks
Labrum or upper lip	It is with a ventral surface called the epipharynx
Hypopharynx	A tongue like structure
Mandibles or jaws	Crush food and may be used for defense
Maxillae (Maxilla)	Hold the food
Labium or lower lip	

❖ Types:

- **Chewing and biting parts (Mandibulate):** Used to chew holes in leaves, bore in stems. These types of mouth parts are present in cockroach, grasshopper, cricket, beetles and earwigs.
- **Piercing and sucking type:**
 - ✓ Examples: **mosquitoes, stink bugs, etc.**
 - ✓ True bugs (Hemiptera), thrips (Thysanoptera), fleas (Siphonaptera) and sucking lice (Psocodea: Anoplura).

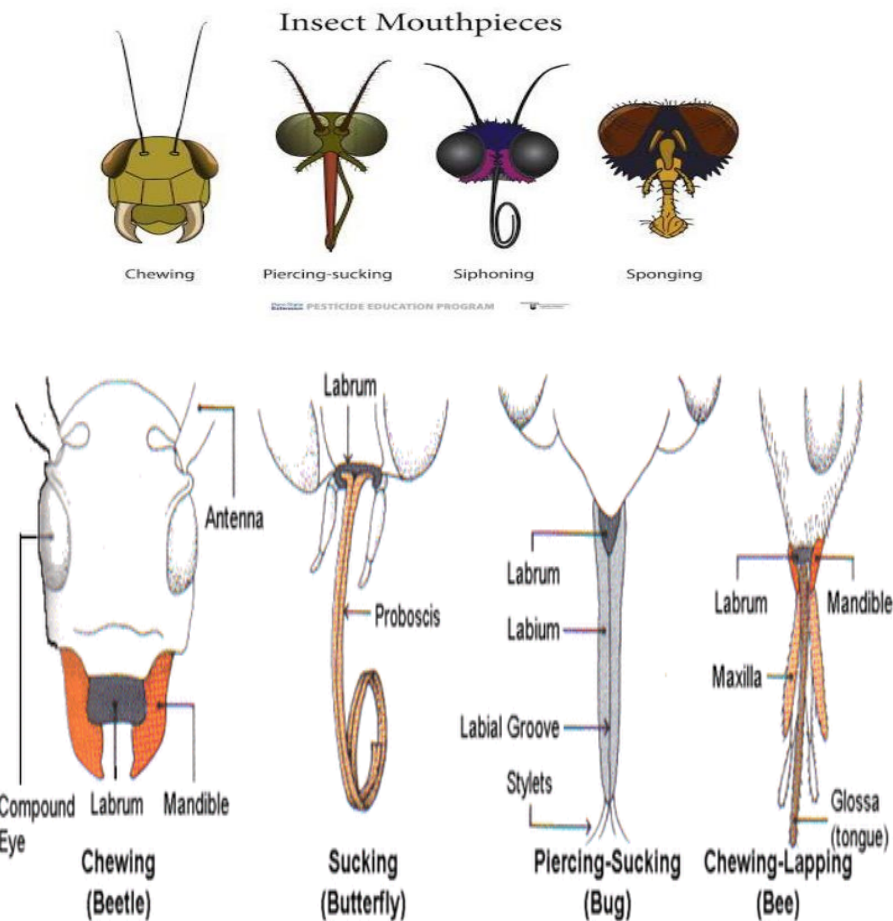
- ✓ In each order different mouth part components form needle like stylets capable of piercing the plant or animal tissues upon which the insect feeds.
- ✓ Bugs have extremely long thin paired mandibular and maxillary stylets, which fit together to form a flexible stylet bundle containing a food canal and a salivary canal.
- **Chewing and piercing-sucking:** Rasp (scrap) surfaces of leave, suck up sap. Example: thrips, Thrips have 3 stylets- paired maxillary stylets (Laciniae) plus the left mandibular one.
- **Sponging Type:**
 - ✓ Modified for liquids or solid foods.
 - ✓ Solid foods must be dissolved by salivary secretions.
 - ✓ **Example: House fly**
- **Siphoning Type:**
 - ✓ Mouthparts form a sucking tube (proboscis).
 - ✓ Modified for uptake of nectar/liquids.
 - ✓ Coiled beneath head when not in use.
 - ✓ **Examples: Butterflies and moths**
- **Rasping-Lapping Type:**
 - ✓ Modified to use liquid or semi-liquid foods.
 - ✓ Some mouthpart components function for chewing
- **Mold wax**
- **Grasping prey**
- **Cutting flowers**
 - ✓ Other components form the proboscis 'lapping' surface.
 - ✓ **Lapping** is a mode of feeding in which liquid or semi-liquid food adhering to a protrusible organ or tongue is transferred from substrate to mouth.

❖ TYPES OF MOUTH PARTS:

Types of Mouth Parts

Examples

Biting and Chewing	Locust, Grasshopper, Cricket
Piercing & sucking type	Mosquito, Aphid, Bugs, Leaf hopper
Sponging Type	Housefly
Siphoning type	Butterfly
Rasping and sucking	Thrips
Rasping and lapping	Honeybee



❖ **Thorax:**

- **Divided** into 3 regions.

- 1. Prothorax:** Bears 1 pair of legs
- 2. Mesothorax:** Bears 1 pair of legs, 1 pair of wings
- 3. Metathorax:** Bears 1 pair of legs, 1 pair of wings

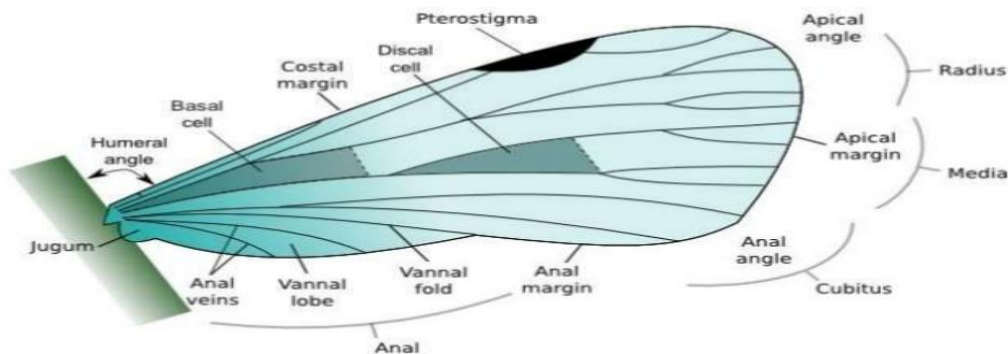
- ❖ **Wings:** Number of wings varies by species. In general, have 2 pairs.

- **Mesothoracic wing:** Forewing
- **Metathoracic wing:** Hindwing

❖ Functions

- Locomotion.
- Protection.
- Camouflage

❖ Wing modifications:



❖ TYPE OF WINGS:

Types	Remarks	Examples
Halters (Halter)	Knob-like reduced hind wings of Diptera	House fly
Elytra (Elytron)	Hardened, protective forewings of Coleoptera	Beetle and weevils
Hemelytra	Half-hardened, half-membranous forewings of Hemiptera	Red cotton bug
Membranous wings		Dragon flies, bees, wasp, grasshopper
Fringed wings	Modified wing structure of the Thysanoptera	Thrips
Scales and hairs		Lepidoptera, Trichoptera, some Diptera Moth and butterflies
Tegmina		Cockroach

❖ Legs

- **Three** pairs of true legs.
- **6 basic segments of the leg.**
- **Adapted** for various functions.

Coxa	Trochanter	Femur
Tibia	Tarsus	Pretarsus

❖ Modifications:

- **Ambulatorial/Gressorial or walking Type:** Usually adapted for walking: e.g., cockroach & bugs.
- **Cursorial or running type:** It is almost similar to walking type of legs but is differentiated by the tarsus which is comparatively longer and touches the ground while running. e.g., Earwigs, Ants.
- **Scansorial or clinging type:** e.g., head louse.
- **Saltatorial or Jumping type:** Eg grass- hoppers, crickets and flea beetle, These muscles help in jumping the insect by their repeated contraction.
- **Stridulatorial or sound producing:** These legs are typically adapted for producing sound wherein the femur of hind leg of male grasshopper or cricket is provided with the row of pegs (file) on its inner side.
- **Fossorial or Digging type:** In certain insects the forelegs are modified for digging purpose. e.g., Male cricket, Nymphs of cicada, grubs of Scarabaeids and carabids.
- **Natatorial or swimming type:** These legs are found in insects living in water and help them in swimming. E.g., Giant water Bug.
- **Foragial or pollen collecting type:** This type of modification is found in the legs of worker honeybees which is mainly adapted for carrying the pollen from the flowers.
- **Raptorial or grasping type:** Such legs are adapted for catching the prey and are found in mantis.
- **Sticking type:** These are also termed as adhesive type of legs which are generally found in house fly.

- ❖ **Abdomen:** Each abdominal segment is made up of only two sclerites namely dorsal body plate (Tergum) and ventral body plate (Sternum).
- ❖ **Spiracles:** These are openings involved in respiration. These are located on each side of abdomen.
- ❖ **Cerci:** Sensory organs.
- ❖ **Aedeagus:** Male copulatory organ.
- ❖ **Ovipositor:** Egg-laying structure

TERMINOLOGY IN ENTOMOLOGY:

- ❖ **Economic damage** was originally defined as the amount of injury which will justify the cost of artificial control measures.
- ❖ **Injury** is defined as the physical harm or destruction to a valued commodity caused by the presence or activities of a pest (e.g., consuming leaves, tunneling in wood, feeding on blood, etc.).
- ❖ **Damage** is the monetary value lost to the commodity as a result of injury by the pest (e.g., spoilage, reduction in yield, loss of quality, etc.). Any level of pest infestation causes injury, but not all levels of injury cause damage. Plants often tolerate small injuries with no apparent damage and sometimes even over-compensate by channelling more energy or resources into growth terminals or fruiting structures. A low level of injury may not cause enough damage to justify the time or expense of pest control operations.
- ❖ **Incubation period:** The incubation period is the time between egg laying and egg hatch.
- ❖ **Larvae:** juvenile form of an insect. Larvae that undergo incomplete metamorphosis are called **nymph**.
- ❖ **Pupa:** life stage of some insects undergoing transformation.
- ❖ **Nymph:** immature form of some invertebrates which undergo metamorphosis to reach adult stage.
- ❖ **Grub:** The larva of some insects especially coleopterans, hymenopterans, neuropterans, etc.
- ❖ **Maggot:** Maggots are the larvae of flies.
- ❖ **Looper:** Larva of some lepidopteran insects having two pairs of abdominal legs (prolegs) on 6th and 10th segment.
- ❖ **Overwintering:** Overwintering is to pass through or wait out the winter season, or to pass through that period of the year when “winter” conditions make normal activity or even survival difficult or near impossible.
- ❖ **Adult:** mature stage of an insect.
- ❖ **Honey dew:** Honeydew is a sugar-rich sticky liquid, secreted by aphids and some scale insects as they feed on plant sap. When their mouthpart penetrates the phloem, the sugary, high-pressure liquid is forced out of the gut's terminal opening.
- ❖ **Parasite:** A **parasite** is an organism that lives and feeds in or on a larger host.

❖ **Parasitoid:** Insect parasite of an arthropod which is parasitic during immature stages, but the adult is free living. Parasitoid is associated with one host and generally kills its host.

❖ HISTORY

- In **1859**, V. M. Stern and colleagues formally proposed the concept.
- They **developed** the ideas of economic damage, economic injury level, and economic threshold, collectively called the economic-injury level (EIL) concept. Although this concept was originally proposed in 1959, some of the ideas expressed had been discussed years earlier (1934) by W.D. Pierce.

❖ Economic damage and damage Boundary

- **Economic damage** was originally defined as the amount of injury which will justify the cost of artificial control measures.
- **Damage boundary**, defined as the lowest level of injury where damage can be measured

CATEGORIES OF PESTS

- ❖ **Key pest:** These are most severe and damaging pest. GEP lies well above the DB and EIL. E.g., Cotton boll worm, DBM, gram pod borer.
- ❖ **Major pest:** GEP is close to EIL but economically damage avoided by timely interventions. Sucking pests of cotton and rice.
- ❖ **Minor pest:** GEP lies below EIL and Damage boundary. Under favorable environmental condition the population may cross EIL and DB for a short interval and a single application of insecticides is usually enough to prevent damage. E.g., Thrips, mites and sugar cane mealy bug.
- ❖ **Regular pest:** Affect specific seasonal crops like cereals, pulses, fruits, passes through many generations during the crop period." E.g., rice stem borer
- ❖ **Sporadic pest:** Population of these insect is usually negligible but in certain year under favorable environmental condition they appear epidemic and crossing many times over EIL and DB. Required suitable cultural tactics to reduce population of these pests. White grub, hairy caterpillar, cut worm, grasshopper.
- ❖ **Potential pest:** These insects are presently not causing any economic damage. Therefore, as such should not be labeled as pests. The GEP lies below the DB and doesn't cross EIL even under favorable conditions. But any change in ecosystem may push their GEP higher and there is a danger of economic damage from these pests if control operations against the other categories of pests are undertaken in an indiscriminate manner. Spodoptera litura on cotton/Soybean. Army worm on wheat is example of potential pest.

APICULTURE:

- ❖ Beekeeping is defined as maintenance of beehives by humans.
- ❖ Honeybees are considered as important pollinators.
 - ❖ **Different species of honeybees**
 - There are four well known species of true honeybees (belonging to genus *Apis*) in the world:
 1. Rock bee, *Apis dorsata*
 2. Little bee, *A. florea*
 3. Asian bee, *A. cerana*
 4. European bee, *A. mellifera*
 - ❖ **Characteristics of four well known species of honeybees**

	<i>Apis dorsata</i>	<i>Apis florea</i>
Nesting	Open nesting. Builds single large comb (ca 1m ²) attached to branches of trees or rocks etc.	Open nesting. Builds single small comb (ca size of palm of hand) fixed to branches of bushes.
Distribution in India	Found in plains as well as hills up to 1600 meters above sea level. Highly migratory.	Found in plains up to 300 metres above sea level. Highly migratory.
Size	Biggest honeybee (16-18mm)	Smallest <i>Apis</i> bee (9-10mm)
Average honey yield per colony/year	40 kg (wild bees; cannot be domesticated)	500 g (wild bees; cannot be domesticated)
Method of Honey extraction	By squeezing (unhygienic)	By squeezing (unhygienic)
Number of cells/10cm comb (worker cells)	18-19	32-36

	<i>Apis cerana</i>	<i>Apis mellifera</i>
Nesting	Cavity nesting. Builds many parallel combs in cavities of tree trunks, hollows of rocks, poles and	Cavity nesting and similar in habits to <i>Apis cerana</i> and builds parallel combs.

	other covered places	
Distribution in India	Found throughout India	Exotic bee to India. Introduced successfully in 1962.
Size	Medium size (14-15mm)	Medium size (14-16mm)
Average honey yield per colony/year	3 kg (Hive bees; can be domesticated)	15 kg Hive bees; can be domesticated
Method of Honey extraction	By centrifugal honey extractor from the hived bees (hygienic).	By centrifugal honey extractor from the hived bees (hygienic)
Number of cells/10cm comb (worker cells)	21-25	17-19

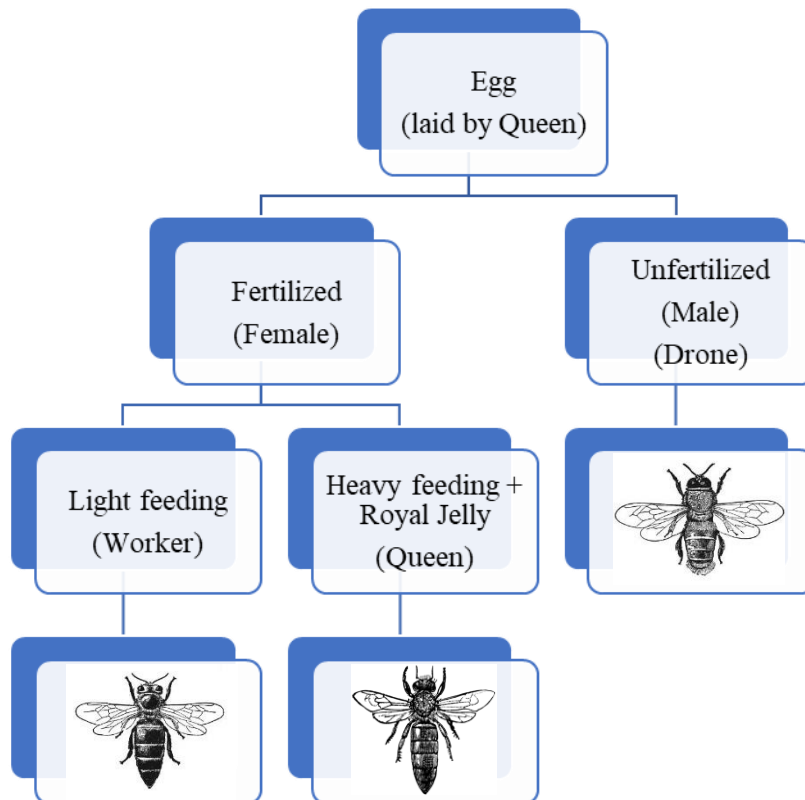
❖ **Like any insect, body of honeybee can be distinguished in to three parts:**

1. Head
2. Thorax
3. Abdomen

1. Head

- Bears a pair of geniculate antennae.
- Mouth parts of worker bees are modified for sucking and lapping.
- Inside the head there are long coiled strings of small lobes known as hypopharyngeal glands which secrete glandular food known **as royal jelly** that is fed to queen and young larvae.
- **Prothoracic legs serve as antenna cleaner.**
- On Mesothoracic legs, **bushy tarsi serve as brushes for cleaning of thorax.**
- In worker bees, smooth somewhat concave **outer surface of hind tibia is fringed with long curved hairs and forms pollen basket or corbicula.**
- In workers **egg laying apparatus (ovipositor) is modified into sting.**
- Queen uses ovipositor for **egg laying and stinging rival queen.**
- Digestive system is unique in having oesophagus with expanded **honey stomach which stores the collected nectar.**
- Sperms are stored in the queen in a sac like structure known as **spermatheca**. The stored sperms are utilized by queen throughout her lifetime as she does not go for mating once starts egg laying.

- A normal colony, during active season is composed of 3 kinds of individuals: one queen, thousands of workers (10000 to 30000 or even more) and few hundreds of drones, which vary in size. **In addition, each colony has different developmental stages viz eggs, larvae and pupae which are collectively known as brood.**



❖ Queen:

- Only one queen is found in a colony except under supersedure or swarming instinct.
- She is mother of whole colony producing workers and drones and is the only perfectly developed female member of the colony.
- Her function is to lay eggs. She does not have motherly instinct or ability to feed the brood. She is fed lavishly by a large number of nurse bees with highly nutritious food known as royal jelly.
- A good queen can lay **1500-2000** eggs per day.
- A laying queen is the **longest bee** in the colony.
- **Fertilized eggs produce workers (also queens) and unfertilized eggs produce drones.**
- A good, mated queen may work satisfactorily for 2 or more years, although queens can live eight years or longer.

- The virgin queen mates with a number of drones (5-7) within 5-10 days of emergence in the air (not inside the hive) and spermatozoa are stored in spermathecal. Stored sperms are utilized to fertilize eggs throughout her life till exhausted.

❖ **Worker:**

- Workers are imperfect females. They are unable to mate though they may start egg laying if a colony remains queen less for long period.
- The workers perform all the useful work in the colony.

❖ **Laying workers:** Under queen less conditions for a long duration, ovaries of some of the workers start developing and they can lay even eggs but since these are unfertilized, give rise to only drones.

❖ **Drone:**

- Drones do not perform any duty inside the hive.
- A colony rears and tolerates the drones only during breeding season.
- The **sole function** of a drone is to mate once which costs him his life.
- Maximum life of drone honeybee in summer is 59 days

❖ **Life cycle:**

- Queen Deposits egg at the base of cell and fastens with mucilaginous secretion.
- After 3 days egg hatches and workers provide pearly white food in which “C” shaped larva floats.
- Cell is sealed when larva is fully grown. In the sealed cell it turns into pupa from which adult emerges.
- Larva sheds skin five times during development.
- The sealed cells containing worker and drone brood and honey can be differentiated based on appearance.

Caste	Egg period (days)		Larval Stage(days)		Pupal Stage (days)		Total (days)	
	A. cerana	A. mellifera	A. cerana	A. mellifera	A. cerana	A. mellifera	A. cerana	A. mellifera
Queen	3	3	5	5	7-8	8	15-16	16
Worker	3	3	4-5	5	11-12	12-13	18-20	21
Drone	3	3	7	7	14	14	24	24

- ❖ **Trophallaxis** is food transmission (exchange of food) which is common between workers and also from workers to queen and drones. It is a sort of communication regarding availability of food and water and also a medium for transfer of pheromone.

- ❖ **Beehive:** L.L. Langstroth discovered the principle of bee space in 1851 in the U.S.A. The bee space measures 9.52 mm for *A. mellifera*, and this was modified for *A. cerana* to be between 7 and 9 mm.

TERMS RELATED WITH APICULTURE

❖ **Swarming**

- It is the natural instinct of honeybees to reproduce its colonies.
- By swarming strong colonies are divided naturally.
- It occurs mostly when the colony population is at its peak.

❖ **Causes:**

- Overcrowding and lack of ventilation.
- Presence of old queen.
- Sudden honey flow.
- Lack of space for egg laying and honey storage

❖ **Absconding:**

- It is the **total desertion of colony** from its nest due to incidence of disease/pest attack, too much interference by human beings or robbing of honey by bees from other colonies.
- Proper hive management can prevent it.

❖ **Trophallaxis:**

- It is food transmission (exchange of food).
- Common between workers and also from workers to queen and drones.
- Communication regarding availability of food and water and a medium for transfer of pheromone.

❖ **Indian bee:** *Apis cerana indica*.

❖ **European bees [Italian bees]:** *Apis mellifera* (It is known to be resistant to TSBV)

❖ ***A. m. capensis* cape bee:** unique thing, laying workers treated as queen (pseudo-queen) but do not mate.

SERICULTURE

- ❖ Sericulture, or silk farming, is the cultivation of silkworms to produce silk.
- ❖ The silkworm filament secreted from the salivary gland of the larva helps us in producing silk.

Types of silk	Scientific name	Host plant
Mulberry silk	<i>Bombyx mori</i>	Mulberry
Eri silk	<i>Samia cynthia</i>	Castor
Tasar silk	<i>Antheraea</i> spp.	Arjun, Ber, Saal
Muga silk	<i>Antheraea assama</i>	Som, Sualu

- ❖ Female silk moths will lay up to 300 eggs at a time.
- ❖ About 2,000 – 5,000 cocoons are needed to make 450g of silk.
- ❖ The silkworm caterpillar builds its cocoon by producing and surrounding itself with a long, continuous fiber, or filament.
- ❖ Liquid secretions from two large glands within the insect emerge from the spinneret, a single exit tube in the head, hardening upon exposure to air and forming twin filaments composed of fibroin, a protein material.
- ❖ A second pair of glands secretes sericin, a gummy substance that cements the two filaments together.
- ❖ Because an emerging moth would break the cocoon filament, the larva is killed in the cocoon by steam or hot air at the chrysalis stage.
- ❖ Silk containing **sericin is called raw silk**

LAC CULTURE

- ❖ Lac is the resinous secretion of lac insects.
- ❖ It involves proper care of host plants, regular pruning of host plants, propagation, collection and processing of lac.
- ❖ **Scientific name:** *Laccifer lacca*
- ❖ **Order:** Hemiptera
- ❖ **Host plants:** Kusum, Ranjeeni (Khair) and Ber (Plum) trees. The insects live upon plant juice.
- ❖ Lac is a natural resin of animal origin.
- ❖ It is secreted by an insect, known as lac-insect.

- ❖ In order to obtain lac, these insects are cultured, and the technique is called lac-culture.
- ❖ It involves proper care of host plants, regular pruning of host plants, propagation, collection and processing of lac.
- ❖ The insects live as a parasite, feeding on the sap of certain trees and shrubs.
- ❖ Mass movement of larvae from female cell to the new off-shoots of host plant, is termed as “swarming”
- ❖ The lac insects repeat its life cycle twice in a year.
- ❖ There are actually four lac crops since the lac insects behave in two ways either they develop on Kusum plants or develop on plants other than Kusum.
- ❖ The lac which grows on Non-Kusum plants is called as “Ranjeem lac,” and which grows on Kusum plant is called as “Kusumi lac.
- ❖ Lac cut from the host plant is called as “stick lac”.
- ❖ Lac can be scraped from the twigs before or after the emergence of larvae. If it is used for manufacturing before the emergence of larvae, the type of lac produced is called as “**Ari lac**” and if it is used for manufacturing purpose after swarming of larvae has occurred, the lac is said to be **Phunki lac**”.
- ❖ The scraping of lac from twig is done by knife, after which they should not be exposed to sun. The scraped lac is grinded in hard stone mills. The unnecessary materials are sorted out. In order to remove the finer particles of dirt and colour, this lac is washed repeatedly with cold water.
- ❖ Now at this stage it is called as “**Seed lac**” and is exposed to sun for drying.
- ❖ Seed lac is now subjected to the melting process.
- ❖ The melted lac is sieved through cloth and is given the final shape by molding. The final form of lac is called “**Shellac**”
- ❖ **Usage:**
 - Gramophone record
 - Floor polishes
 - Paints
 - Electrical goods
 - Sealing wax
 - Printing ink

SOME IMPORTANT POLYPHAGOUS PESTS

- Aphid, Thrips, Whiteflies, Borer- *Helicoverpa armigera* , Red spider mite, Red hairy caterpillar, Locust

❖ Aphids:

- Known also as greenfly, blackfly or plant lice, aphids are minute plant-feeding insects.
- Important natural enemies include the predatory ladybugs/ladybirds/ladybeetles, and lacewings.
- Aphids are tiny, pear-shaped, sap-sucking pests, appearing in the on your plants' tender new leaves. They leave behind a secretion that attracts ants and promotes mold growth.

❖ Thrips:

- Thrips (order Thysanoptera) are minute (mostly 1 mm long or less), slender insects with fringed wings and unique asymmetrical mouthparts.
- Thrips can cause damage during feeding.
- Nymph and adult lacerates leaves from the under surface of the leaves and flower buds.
- As a result, white streaks appear on the infested leaves. Leaves show brown patches and get distorted, finally wither and drop down. Infested flowers do not open, flowers fade and drop down prematurely.

❖ Red spider mite

- Larvae, nymphs and adults cause damage to the host plant by feeding on plant sap.
- They mainly occur on the underside of leaves where they pierce the cells and suck out the contents.
- The empty dead cells become yellow, and in many plants the damage can also be seen on the upper surface of leaves as small yellow dots.
- The destruction of cells results in reduced photosynthesis, increased transpiration and reduced plant growth.
- As damage increases, whole leaves turn yellow, and as more cell sap is removed, the leaf, and eventually the whole plant, may die.
- The nymphs and adults also produce webs, and plants can get completely covered with such webs in which the mites live.
- The webbing and spotting on the leaves affects the appearance of the crop

❖ Red hairy caterpillar (*Amsacta moorei*)

- **Family:** Arctiidae
- **Order:** Lepidoptera.
- Body of caterpillars is covered with the numerous long hairs arising from the fleshly tubers.
- The entire abdomen is scarlet red there are black bands and dots on the abdomen.
- Feed on gregariously and as they grow older moult 6 times.
- During a severe attack, the caterpillars in bands destroying fields after field.

❖ **White grub** (*Holotrichia consanguinea*)

- **Family:** Melolonthidae
- **Order:** Coleoptera.
- Damage done by grub in field crops and by adults in trees.
- The damage caused by grub is so much so that entire stand crop is destroyed thereby necessitating the resowing in field.
- **Designated as a 'National pest'.**
- The plant damaged by the grub gives a wilted appearance and finally dries out.
- The full-grown grubs move down deeper in the soil in search of moisture & for pupation.
- After pre monsoon rain/ first shower of monsoon the adult comes out from soil and cause damage to trees during night.
- Eggs are laid in singly in loose moist sandy or sandy loam soil on the onset of monsoon.
- Both grubs and adults are damaging.
- Grubs feed on underground plant parts of various crops.
- Older second instar and third instar grubs are more damaging.
- Due to concealed feeding white grubs generally remain unnoticed and at harvest a large number of tubers are found infested/damaged.

- White grubs are also found to feed on the roots of horticultural/ forest nurseries and some ornamental plants.
- Adult beetles feed on the foliage of many trees.
- There are three larval instars.
- Beetles emerge during monsoon.
- Lay eggs in soil.
- Eggs are laid singly in 2-7 installments.
- Each female lays 4-40 eggs in its life span.
- The pest passes winter as grub in earthen cells.
- Pupate in cells during April – May.

❖ **Locust:**

- **Family:** Acrididae
- **Order:** Orthoptera
- Locusts are the swarming phase of short-horned grasshoppers.
- They can breed rapidly under suitable conditions and subsequently become gregarious and migratory.
- Nymphs form bands and adults' swarms.
- They can travel great distances, rapidly stripping fields and greatly damaging crops.
- The origin and apparent extinction of certain species of locust (some of which were 150 mm in length) is unclear.
- Adult and nymphs cause the damage.
- Gregarious and voracious feeders.
- Eat up the entire vegetation.
- In spite of some expensive control measures, the damage caused to crops by locusts during 1926-31 cycle was estimated to 100 million.
- They also climb on walls, invade kitchens, storerooms thus causing nuisance.

- Eggs are laid in soil in egg pods.
 - Each female can lay up to 11 egg pods, each pod containing up to 120 eggs.
 - Before egg laying each female with the help of its ovipositor, bores a 5-10 cm deep hole into the loose soil.
- (a) **Bombay locust (*Pantanga succinct L.*):** Mostly confined to Gujarat and Tamil Nadu. Its breed during monsoon in the Western Ghats and has only one brood in a year.
- (b) **Migratory locust (*Locusta migratoria*):** Mainly occurs in Rajasthan and Gujarat. Its breed twice in a year: winter -spring breeding occurs in Pakistan and summer-monsoon breeding in Rajasthan and Gujarat. It may, have many broods a year.
- (c) **Desert locust (*Schistocera gregarea*):** This is most destructive of all locust The adult of this species occurs **in two phases**:
- (i) **Solitary phase (*solitaria*):** wherein the insects remain scattered
 - (ii) **Gregarious phase (*gregarea*):** wherein the insects congregate as nymphs and adults.
- **Life -history:**
- ✓ Mating occurs between sexually mature yellow adults.
 - ✓ Egg -laying starts after 8-24 hours of mating, a single female can lay up to 500 eggs in about 5 egg -pods or capsule.
 - ✓ Female inserts her long ovipositor in sandy soil to bore a hole 2-4 inches deep.
 - ✓ The summer breeding occurs during June —September (monsoon) in this quiescent eggs are hatched out and nymphal period is about 12-15 days, within this period emerges in five nymphal instars.
 - ✓ Within the period of one-month nymphal stage emerges into adult.
- **Damage:** Both nymphs and adults cause damage
- **Host plants:** Mainly pearl millet, sorghum, maize but being Polyphagous nature, it's eat on all kind of vegetation except a few namely, aak, neem, datura, Jamun and sheesham.
- **Management:**
- ✓ Eggs can be destroyed by deep summer ploughing or flooding.

- ✓ The nymphs are most vulnerable stage for management point of view.
- ✓ Because at second and third nymphal stage is key stage where pest population can be checked.
- ✓ A trench can be dug around the breeding field of locust so that when nymphs emerge, they can be buried under soil, or methyl parathion 2% can be dusted to kill the emerging nymphs.
- ✓ When infestation starts on field crops or vegetation around the field duts, methyl parathion 2% or Malathion 5% @ 25 kg/ha on field crops including field border and vegetation of non-cropped area.

❖ **Termites** (*Odontotermes obesus*,)

- **Family:** Termitidae
- **Order:** Isoptera.
- Also known as **white ants**.
- Termite lives in social colony in underground nests make earthen mounds (termatoria).

A. Reproductive castes:

- (i) **Colonizing individuals:** Winged individuals of both sexes and are produced in large number during rainy season. After mating they cast off the wings and start a new colony.
- (ii) **Queen:** Only perfectly developed female in the colony. It measures about 5.0-7.5 cm in length, and she is known as a phenomenon, '**Egg laying machine**'. It is laying one egg/second or 70,000-80,000 eggs in 24 hours. Queen fed by workers; choicest food lives in "royal chamber" which situated in center of nest (0.5 m depth).
- (iii) **King:** It developed from unfertilized eggs. It lives with queen in royal chamber & smaller than queen and mates with queen from time to time.
- (iv) **Complementary caste:** Short winged or wingless of both sexes lead a subterranean life. They replaced in place of queen & king after ultimately death of them.

(v) **Sterile caste:**

a. Workers

b. Soldier

- **Management:**

- ✓ When a colony is established, it is not so easy to eradicate; only sure method is to reach the centre of nest and kill the queen and the complementary forms.
- ✓ Before sowing granular application of Regent (Fipronil 5 % SC) @ 20 kg /ha.
- ✓ Should not be use of undecomposed green manure and FYM at termite susceptible field.
- ✓ Seed treats with Chlorpyriphos 20 EC @ 450m1/100 kg of seeds in wheat.
- ✓ In standing crop application of Chlorpyriphos 20 EC @ 4 liter/ha

❖ Cutworms

➤ Important species:

Agrostis segetum	A. epsilon
A. flammata	A. Spinifera

- **Family:** Noctuidae
- **Order:** Lepidoptera
- **Caterpillars** are damaging, Cut the seedling at ground level. Drag the seedling into the soil. Reduce the plant strand. Sometime replanting is required. Caterpillars also nibble the tubers of plants.

METHODS OF PEST CONTROL

- Any factor that is capable of making life hard for the insect that will repel or interfere with its feeding, mating, reproduction or dispersal can be taken as a method of insect control in its broadest application.

❖ Different methods of pest control / Components or tools of IPM are

- Cultural methods
- Mechanical methods
- Physical methods
- Biological methods
- Legislative methods

- Host plant resistance
- Chemical methods

❖ Cultural Methods of Pest Control

- The manipulation of cultural practices at an appropriate time for reducing or avoiding pest damage to crops is known as cultural control.
 - The cultural practices make the environment less favorable for the pests and or more favorable for its natural enemies.
 - It is the cheapest of all methods.
- Proper preparatory cultivation:** E.g., Pupae of moths, roots grubs etc.
 - Clean cultivation:** Paddy gall fly *Orseolia oryzae* breeds on grasses such as *Panicum* sp., *Cynodon dactylon* etc. Fruit sucking moth larvae *Eudocima ancilla* on weeds of Menispermaceae.
 - Systematic cutting and removal of infested parts:** Eg. Removal of sugarcane shoots affected by borers, Cutting and removal of infested parts of brinjal attacked by *Leucinodes orbonalis*. Clipping of tips of rice seedlings before transplanting eliminate the egg masses of stem borer. Clipping of leaf lets in coconut reduces the black headed caterpillar.
 - Changes in the system of cultivation:** Change of banana from perennial to annual crop reduced the infestation of banana rhizome weevil *Cosmopolitus sordidus* in addition to giving increased yields. Avoiding ratoon redgram crop during offseason helps in reducing the carryover of pod fly *Melangromyza obtusa* and eriophyid mite *Aceria cajani*
 - Crop rotation:** Lady's finger followed by cotton will suffer from increased infestation of pests. Cereals followed by pulses. Cotton should be rotated with non-hosts like ragi, maize, rice to minimize the incidence of insect pests.
 - Mixed cropping:** Intended for getting some return when one crop is attacked, the other escapes. E.g., Garden peas and Sunhemp.
 - Growing resistant varieties:** Certain varieties resists pest attack. Eg: GEB-24 and MTU-5249 resistance to paddy BPH, Surekha variety to gall midge, TKM -6 and Ratna for stem borer.

❖ Cultural practices specially adopted for certain pests:

1. Adjusting planting or sowing or harvesting times to avoid certain pests: E.g. Early planting of paddy in *kharif* and late planting in *rabi* minimize the infestation of rice stem borer.

2. Trap cropping:

Trap Crop	Main Crop	Insect Pest
Castor	Chillies	Tobacco caterpillar, <i>Spodoptera litura</i>
Tomato	Citrus	Fruit sucking moths, <i>Otheris spp</i>
Marigold	Cotton	American bollworm, <i>Helicoverpa armigera</i>

3. Trimming field buds: Grasshopper eggs, which are laid in field bunds are destroyed by trimming field bunds.

4. Flooding the field: Flooding of fields is recommended for reducing the attack of cutworms, army worms, termites, root grubs etc., Eg: For cutworms like paddy swarming caterpillar (*Spodoptera mauritinana* and *S. exiqua*)

5. Draining the fields: In case of paddy case worm *Nymphula depunctalis* which travel from plant to plant via water. it can be eliminated by draining or drying the field.

6. Alley ways: Formation of alley ways for every 2 m in rice field reduces the BPH *Nilaparvata lugens*

❖ **Legislative / Legal / Regulatory Methods of Pest Control:** Some exotic pests introduced into our country.

Common Name	Scientific Name
Potato tuber moth	<i>Pthorimea operculella</i>
Cotton cushiony scale	<i>Icerya purchasi</i>
Woolly aphid on apple	<i>Eriosoma lanigerum</i>
San Jose scale	<i>Quadraspidiotus perniciosus</i> ,
Golden cyst nematode	<i>Globodera rostochinesis</i>
Giant African snail	<i>Achatina fulica</i> (Predatory snail <i>Eugladina rosea</i>)
Serpentine leaf miner	<i>Liriomyza trifolii</i>
Spiralling whitefly	<i>Alerodicus dispersus</i>
Coconut mite	<i>Aceria guerreoronis</i>

❖ Quarantine:

- The word quarantine is derived from Latin word *Quarantum* which means ‘forty (40)’.
- The first Quarantine Act in USA came into operation in **1905**.
- While GOI passed an Act in 1914 entitled “**Destructive Insect and Pests Act of 1914**”_to prevent the introduction of any insect, fungus or other pests into our country.
- The **Directorate of Plant Protection Quarantine and Storage (DPPQS)** was established in Faridabad in 1946.
- These operates under the provisions made under the “**Destructive Insect and Pests Act of 1914**”.

- NBPGR, New Delhi for quarantine processing of all germplasm under exchange for research purposes, FRI, Dehradun for forest plants and Botanical Survey of India (BSI), Kolkata for ornamental plants to enforce quarantine laws.
- Import of potatoes from areas known to be infected with wart disease or golden cyst nematode is totally prohibited into our country.
- The consignment should accompany with the certificate issued by the Officers of agriculture department of the exporting country. This certificate is called as **‘Phytosanitary certificate’**.
- The Insecticide Act, 1968 has been enforced on 2nd September 1968 by the GOI to regulate the import, manufacture, sale, transport and distribution and use of insecticides.
- The government of India also constituted the Central Insecticide Board (CIB) to advise the state and central governments as per this act. The insecticide rules of 1971 framed under the Insecticides Act 1968 had come into force in 1971.
- ❖ **Mechanical Methods of Pest Control:** Reduction or suppression of insect pest population by means of manual devices or labour
 - (a) Hand picking and collection with hand nets and killing insects
 - (b) Provision of preventive barriers:
 - (c) **Other mechanical methods:**
 - Extraction of adult Rhinoceros beetle (*Oryctes rhinoceros*) from the crown of coconut trees using an arrow headed rod/hook.
 - Construction of rat proof godowns.
 - Use of an alkathene band around the tree trunks of mango to check the migration of first instar nymphs of mealybugs and red ants.
 - Sticky bands around tree trunks against red tree ant (*Oecophylla samaragdina*).
 - Systematic shaking of root grub adults harbored trees during evening hours to dislodge and destroy by dumping in fire.
 - Shaking of red gram plants to collect and destroy later instars of *Helicoverpa armigera*
 - Shaking the trees and bushes by which the insects fall to the ground and they can be collected.
 - Sieving and winnowing against stored grain pests.
 - Using mosquito nets fly proof cages etc.

❖ **Physical Methods of Pest Control:**

- A material called drier, consist of highly porous, finely divided silica gel which when applied abrades the insect cuticle thus encouraging loss of moisture resulting in death. It is mainly used against stored product pests.
 - Kaolinic clay after successive activation with acid and heat can be mixed with stored grain. The clay minerals absorb the lipid layer of the insect cuticle by which the insects lose their body moisture and die due to desiccation.
 - Stored products can be exposed to 55°C for 3 hours to avoid stored product pests.
 - Steam sterilization of soil kills soil insects.
 - Vapour Heat Treatment (VHT): Heated air is saturated with water (>RH 90%) for specified period of 6 to 8 hours for raising pulp temperature to 43-44.5°C in case of mango against fruit flies.
 - Oxygen stress and carbon dioxide concentration
 - **Male sterile technique:** By sterilizing the pupae of screwworm, livestock pest (*Cochliomyia hominivorax*) with radiations, sterile males were obtained. They were released @ 400/sq mile for 7 weeks.
 - Light traps are arranged for attracting the insects: Eg: Most of the moths and beetles.
 - Flame thrower is a compressed air sprayer with kerosene oil for producing flames. There is a lance, which is fitted with a burner. When the burner is heated, the kerosene oil is released, and it turns into flames. Used for burning locust populations, congregation of caterpillars, patches of weeds etc.
- ❖ **Chemical Control:** Control of insects with chemicals is known as chemical control.
 - ❖ **Pesticides:** The term pesticide is used to those chemicals which kill pests, and these pests may include insects, animals, mites, diseases or even weeds.
 - ❖ **Insecticides:** Chemicals which kill insects are called as insecticides. Insecticide may be defined as a substance or mixture of substances intended to kill, repel or otherwise prevent the insects.

DIFFERENT CLASSIFICATIONS OF INSECTICIDES

I. Based on the origin and source of supply

A. Inorganic insecticides:

- Includes arsenate and fluorine compounds.
- Sulphur as acaricides and zinc phosphide as rodenticides.

B. Organic Insecticides:

1. **Insecticides of animal origin:** Nereistoxin isolated from marine annelids, fish oil rosin soap from fishes etc.
2. **Plant Origin insecticides or Botanical insecticides:** Nicotinoids, pyrethroids, Rotenoids etc.
3. **Synthetic organic insecticides:** Organochlorines, Organophosphorus, Carbamate insecticides etc.,
4. **Hydrocarbon oils etc.**

II. Based on the mode of entry of the insecticides into the body of the insect they are groups as

1. **Contact poisons:** E.g., DDT and HCH.
2. **Stomach poisons:** Eg: Calcium arsenate, lead arsenate.

❖ Fumigants:

- A fumigant is a chemical substance which is volatile at ordinary temperatures and sufficiently toxic to the insects.
- Fumigants mostly gain entry into the body of the insect through spiracles in the trachea.
- Commonly used Fumigants and their doses:
 1. Aluminium phosphide, marketed as Celphos tablets used against field rats, groundnut bruchids etc.
 2. Carbon disulphide
 3. EDCT (Ethylene Dichloride Carbon Tetrachloride)
 4. EDB Ethylene dibromide
 5. **SO₂:** By burning sulphur in godowns SO₂ fumes are released.

❖ **Systemic insecticides:** Examples: Methyl demeton, Phosphamidon, Acephate.

❖ **Non systemic insecticides:** E.g., Chlorpyrifos, Cypermethrin, Carbaryl, Deltamethrin.

- Some non-systemic insecticides, however, have ability to move from one surface leaf to the other.
- They are called as **trans laminar insecticides**.
- E.g., **Malathion, Diazinon, Spinosad etc.**

❖ **Biological control**

- Management of a pest by means of another living organism that is encouraged and disseminated by man is called biological control.

❖ **Techniques in biological control:**

- Biological control practices involve three techniques.
 - ✓ Introduction,
 - ✓ Augmentation
 - ✓ Conservation.

❖ **Parasites:**

- **Primary parasite:** A parasite attacking an insect which itself is not a parasite (Beneficial to man.)
- **Secondary parasite:** A hyperparasite attacking a primary parasite (Harmful to man)
- **Tertiary parasite:** A hyperparasite attacking a secondary parasite (Beneficial to man)
- **Quaternary parasite:** A hyperparasite attacking tertiary parasite (Harmful to man)

❖ **Microbial control**

- Microbial control refers to the exploitation of disease-causing organism to reduce the population of insect pest below the damaging levels.
- **Steinhaus (1949)** Coined the term 'Microbial Control' when microbial organisms or other products (toxins) are employed by man for the control of pests on plants, animals or man.

1. Bacteria:

- *Bacillus thuringiensis* (B.t.) is important and is isolated from flour moth, *Ephestia kuhniella* by Berliner (1915)
- Bt known as a bacterial insecticide is now being using by farmers mostly on lepidopterous larvae.

- The entry of the bacteria is by ingestion of the bacteria, which infect the mid gut epithelia cells and enter the hemolymph to sporulate and cause septicemia.
- Available in different formulations (Trade names Thuriocide, Delfin, Bakthane, Biobit, Halt, Dipel etc.).

2. Viruses:

❖ NPV (**Borrellina virus**):

- Nuclear polyhedral virus has been isolated from the order Lepidoptera.
- Among these viruses Baculoviruses (Baculoviridae) are successful in IPM.
- The virus infected dead larvae hanging upside down from plant parts (Tree top disease).
- The cuticle becomes fragile, rupturing easily when touched, discharges liquefied body fluids.
- NPV multiplies in insect body wall, trachea, fat bodies and blood cells.
- The polyhedra are seen in nuclei.
- The polyhedral bodies enlarge in size destroying the host nuclei to get released into the insect body cavity.

3. Fungi:

- The fungal disease occurrence in insects is commonly called as mycosis. Most of the entomopathogenic fungi infect the host through the cuticle.
- Important species are, *Entomophthora*, *Metarhizium*, *Beauveria*, *Nomuraea* and *Verticillium*.
- **Eg:** *Entomophthora grylli* on grasshoppers; *Aspergillus flavus* on *Epilachna* beetles; *Spicaria* sp. on castor whitefly; *Metarhizium anisopliae* (Green muscardine) on Orthoptera, white muscardine, *Beauveria bassiana* on *Leptinotarsa decemlineata*.

4. Entomopathogenic nematodes (EPNs)

- Rhabditids (Rhabditidae) are found to have a symbiotic relationship with the bacteria, forming a disease complex.

- The nematode serves as a vector for the bacterium, which produces a septicemia (sporulation in blood; Milky disease) in the insect body.
- The EPNs *Steinernema* spp and *Heterorhabditis* spp from the families, Steinernematidae and Heterorhabditidae have the mutualistic association with bacteria *Photorhabdus* and *Xenorhabdus* spp., respectively.
- The only stage that survives outside the host is the non – feeding 3rd stage Infective Juvenile (IJ).
- *Rodolia cardinalis* (Vadalia beetle) against cotton cushion scale (*Icerya purchasi*)
- *Trichogramma ostrinae* against European corn borer.
- *Encarsia formosa* against greenhouse white fly.
- The spotted lady beetle (*Coleomegilla maculata*) is also able to feed on the eggs and larvae of the Colorado potato beetle (*Leptinotarsa decemlineata*)
- *Phasmarhabditis hermaphrodita* is a microscopic nematode that kills slugs.
- The bug *Orius insidiosus* has been successfully used against the two-spotted spider mite and the western flower thrips (*Frankliniella occidentalis*)
- *Bacillus thuringiensis* is the most widely applied species of bacteria used for biological control, with at least sub-species used against **Lepidopteran (moth, butterfly).**
- The bacterium *Paenibacillus popilliae* which causes milky spore disease has been found useful in the control of Japanese beetle, killing the larvae.
- *Beauveria bassiana* is mass-produced and used to manage a wide variety of insect pests including whiteflies, thrips, aphids and weevils.

PESTS OF FIELD CROPS

Common name	Scientific name	Family	Order
Yellow stem borer	<i>Tryporyza/Scirpophaga incertulus</i>	Pyralidae	Lepidoptera
Rice hispa	<i>Diclodispa armigera</i>	Chrysomelidae	Coleoptera
Rice gundhi bug	<i>Leptocorisa acuta</i>	Coreidae	Hemiptera
Brown plant hopper	<i>Nilaparvata lugens</i>	Delphinidae	Hemiptera
Rice case worm	<i>Nymphula depunctalis</i>	Pyralidae	Lepidoptera

Rice crop pests

1. Yellow stem borer: *Tryporyza/Scirpophaga incertulus*.

- **Family:** Pyralidae (Lepidoptera)
- Monophagus/absolute pest of rice.
- Full grown larvae hibernate in rice stubbles.
- It pupates inside the plant.
- Full grown larvae measures about 20 mm adults are dirty white or greenish yellow front wings.

❖ **Symptoms of damage:**

- Caterpillar alone is destructive, construct an emergence hole which always located above the water level.
- Caterpillar produce dead hearts on affected plants.
- Plants that are attacked in early stages produce ears devoid of grains and are known as "white ear".
- Basmati varieties suffer heavy damage then coarse varieties.

❖ **Management:**

- Removal & destruction of crop stubbles.

- Since eggs of stem borer are laid near tip of leaf hence, clipping and burning of tips of the seedling as a routine before transplanting.
- Application of Cartap hydrochloride 4G @ 2.5 kg a.i. /ha.

2. **Rice gundhi bug:** *Leptocorisa acuta*

- Family: Coreidae
- Order: Hemiptera
- **Periodic pest of rice.**
- Adult bugs have long legs and are slender about 20 mm long.

❖ **Symptoms of damage:**

- Rice field severely attacked by this pest emit a repugnant smell which gives to this pest the name "Gundhi bug".
- The nymphs and adults suck juice from developing grains in the milky stage, causing incompletely filled panicles or panicles with empty grains.
- Black and brown spots are appearing around the holes made by the bugs on which a sooty mould may develop.
- This pest attacks the rice crop periodically.

❖ **Management:**

- Clean cultivation: Paddy field and surrounding bunds should be kept clean from weeds.
- Adult bugs captured through light traps and destroyed.
- Hand collection of bugs.
- **Resistant variety:** Grow Badshabhog variety.
- Spray Malathion 50 EC @ 675 ml/ha

3. **Brown plant hopper:** *Nilaparvata lugens*

- **Family:** Delphacidae
- Order: Hemiptera
- Serious pest of high yielding varieties of paddy.

❖ Symptoms:

- Both nymph & adult cause damage by sucking cell sap from leaves which turn yellow.
- Under the favorable condition of high humidity, high nitrogen application and no wind, the population increase and "**hopper burn**" is obtained in various localities.
- It transmitted the "grassy stunt" virus disease of rice.

❖ Management:

- Closer spacing should be avoided (follow a spacing of 20 x 15 cm).
- Alternate drying & wetting of field.
- Alley 30 cm wide after every 3 m of rice planting provide proper aeration to crop.
- Spray of 100 ml Imidacloprid (200 SL) with 250 liters of water.
- Multi gene resistant varieties: IR-36, IR-60.

❖ Other pests of rice:

a. Green leaf hopper: *Nephotettix nigropictus/virescens*.

- **Order:** Hemiptera
- **Family:** Cicadellidae
- This is responsible for vector of virus diseases, of which **Tungro** is most serious.

b. White backed plant hopper: *Sogatella furcifera*

- Family: Delphacidae
- Order: Hemiptera
- Cause rust red symptoms

c. Rice gall midge: *Orseolia oryzae*

- **Family:** Cecidomyiidae
- **Order:** Diptera
- Cause "silver shoot & onion leaves" symptoms.

d. Rice thrips: *Stenochaelothrips biformis*

- **Family:** Thripidae
- **Order:** Thysanoptera
- Important pest of rice nursery

e. Rice grasshopper: *Hieroglyphus banian*

- **Family:** Acrididae
- **Order:** Orthoptera
- Damage by nibbling of rice ear head

❖ **IPM Modules for Rice:**

- Hispa: 1-2 adult/hill
- BPH: 5-10 hopper/hill

❖ **Summary points for management**

- Avoiding use of excess nitrogen which increases population of Brown plant hopper (BPH) and leaf folder.
- Alternate wetting & drying for BPH.
- Avoid close planting in case of BPH epidemic area.
- Passing the rope on the crop and draining of water for case worm.
- Using the light traps to monitor BPH, Green plant hopper and stem borer.
- Clipping of seedling tips for rice hispa, thrips & yellow stem borer.
- Use of NSKE (Neem seed kernel extract) 50% or Neem oil for ear head bug

Insect -pest of wheat

Common name	Scientific name	Family	Order	Damaging stage
-------------	-----------------	--------	-------	----------------

Army worm	Mythimna separata	Noctuidae	Lepidoptera	Larvae
Ghujhia weevil	Tanymecus indicus	Curculionidae	Coleoptera	Adult

1. Army worm of wheat: *Mythimna separate*

- **Family:** Noctuidae
- **Order:** Lepidoptera.
- The freshly emerged larvae spin threads from which they suspend themselves in the air and then with the help of air current reach from one plant to another.
- Early-stage feeds on tender leaves in to central whorl of the plants but later caterpillars are able to feed on older leaves and skeletonized them totally.
- In case of severe attack by the army worm whole leaves including the mid ribs are consumed and the field looks as "grazed by the cattle".
- Spray Quinalphos 25 EC @ 500 ml/ha.

2. Ghujhia weevil: *Tanymecus indicus*,

- **Family:** Curculionidae
- **Order:** Coleoptera.
- It is specific pest of wheat nursery.
- Damage caused by adult weevil only.
- They cut the germinating seedlings at the ground level.
- Mainly damage is done in the months of October- November.

3. Other pests of wheat:

a. Termite: *Odontotermes obesus*,

- **Family:** Termitidae
- **Order:** Isoptera.

b. Wheat aphid: *Macrosiphum miscanthi*

- **Family:** Aphididae

- **Order:** Hemiptera.

Insect-pests of maize:

1. Maize/ Sorghum stem borer: *Chilo partellus*

- **Family:** Pyralidae
- **Order:** Lepidoptera.
- Most destructive pest of maize/sorghum.
- Caterpillars infect the crop usually one month after sowing.
- It causes pin holes on leaves or "dead hearts" in stem.
- The infestation may be noticed till harvesting.
- Larvae remains hibernate in maize stubbles.
- ❖ **Management:**
 - Destroy stubbles, weeds & alternate hosts.
 - Application of Phorate 10 G @ 20 kg/ha or Carbofuran 3G @ 25 kg/ha.
 - Release of Cotesia (Apanteles) spp.

Common name	Scientific name	Family	Order
Maize aphid	<i>Rhopalosiphum maydis</i>	Aphididae	Hemiptera
European corn borer. (It doesn't occur in India)	<i>Ostrinia nubilalis</i>	Pyralidae	Lepidoptera
Asian maize borer (present in some part of India)	<i>O. furnacalis</i>		
Pink borer It mainly feeds on maize, rice, and wheat. It causes 'dead heart' on central shoot in wheat.	<i>Sesamia inferens</i>	Noctuidae	Lepidoptera

Insect -Pest of cotton:

Common name	Scientific name	Family	Order
Cotton jassids	<i>Amrasca biguttula</i>	Cicadelidae	Hemiptera
Cotton whitefly	<i>Bemisia tabaci</i>	Aleyrodidae	Hemiptera

Pink boll worm	Pectinophora gossypiella	Gelechiidae	Lepidoptera
Spotted boll worm	Earias insulana vitella	Noctuidae	Lepidoptera

1. Cotton jassids: *Amrasca biguttula*

- **Family:** Cicadelidae
- **Order:** Hemiptera.
- Destructive pest of American cotton.
- Nymphs are wedge shaped and adults are winged form.
- ❖ **Symptoms:**
 - Adult and nymph cause injury to crop is due to loss of sap and probably also due to the injection of toxins.
 - Yellowing and downward rolling/cupping of leaves due to toxins.

2. Cotton whitefly: *Bemisia tabaci*

- **Family:** Aleyrodidae
- **Order:** Hemiptera
- The pale-yellow body is slightly dusted with a white waxy powder.
- They have two pairs of pure white wings and have prominent long hind wings.
- ❖ **Symptoms:**
 - **Damage done by two types.**
 - (a) The vitality of the plants is lowered through the loss of cell sap.
 - (b) The normal photosynthesis is interfered with due to the growth of a sooty mould on the honey dew excreted by the insects.
 - It **transmits** several viral diseases including the leaf curl disease of tobacco, vein curling disease of okra and leaf curl of sesame.
- ❖ **Management:**

- Both jassids and whiteflies can be effectively managed by spraying of Imidacloprid 200 SL @ 75 g a.i. /ha or Thiomethoxam @ 45 g a.i. /ha. I Spray of Dimethoate @ 1 ml/liter of water

3. Pink boll worm: *Pectinophora gossypiella*

- **Family:** Gelechiidae
- **Order:** Lepidoptera
- Most destructive pest of cotton.
- The caterpillars are pink in colour and found inside the flower buds, panicles and the bolls of cotton.

❖ Symptoms:

- Damage done by caterpillar in various ways there is excessive shedding of the fruiting bodies.
- "**Rosette flower**" is characteristic symptoms.
- The total shedding is caused by all the boll worms collectively one half may be due to the attack of pink bollworm.
- **Double seed formation:** The two adjoining seed are joining together within damaged boll by pink boll worm.

❖ Management:

- Growing of Bt. cotton varieties is the effective method for boll worm complex.
- Destruction of cotton sprouts, alternative host plants or burning of the plant debris.
- Deep ploughing (with furrow turning plough) done by the end of February.
- **Larval:** Parasitoid *Bracon greeni*, *Chelonus pectinophorae*.

4. Spotted boll worm: *Earias insulana* / *E. vitelli*

- **Family:** Noctuidae
- **Order:** Lepidoptera.
- It is serious pest of cotton and okra.

- A series of longitudinally black spots on the body with dull green colour and having tiny stout bristles are the main identification mark of spotted boll worm.

❖ Symptoms:

- Cause heavy shedding of fruiting bodies.
- '**Flare square**' (**Flare-up**) is characteristic symptoms.
- In the attacked bolls the lint is spoiled by larval feeding.
- The infected bolls open prematurely and produce poor lint resulting in lower market values.

❖ Other pests of cotton:

Common name	Scientific name	Family	Order
Dusky cotton bug	<i>Oxycarenus laetus</i>	Lycaenidae	Hemiptera
Red cotton bug Red cotton bug stains the lint in ginning factories when it crushed with cotton	<i>Dysdercus singulatus</i>	Pyrrhocoridae	Hemiptera
American boll worm	<i>Helicoverpa armigera</i>	Noctuidae	Lepidoptera
Cotton leaf roller	<i>Sylepta derogata</i>	Pyralidae	Lepidoptera

Insect – pests of Sugarcane

Common name	Scientific name	Family	Order
Top borer	<i>Tryporhyza/ Scirpophaga nivella</i>	Pyralidae	Lepidoptera
Sugarcane leafhopper	<i>Pyrilla perpusilla</i>	Lophopidae	Hemiptera.

1. Top borer: *Tryporhyza/Scirpophaga nivella*

- **Family:** Pyralidae
- **Order:** Lepidoptera.
- Young plants attacked by caterpillars show characteristic reddish streaks on the mid rib.

- They show a number of shoot holes in leaves which ultimately cause dead hearts.
- After cane formation it attacks the terminal portion of canes causing "bunchy tops".

❖ **Management:**

- Collect and destroy moths and egg clusters.
- Cut the attacked shoots at ground level from April to June.
- Application of Carbofuran 3G @ 25 kg/ha or Cartap hydrochloride 4G @ 25 kg/ha in July.

2. Sugar cane leafhopper: *Pyrilla perpusilla*

- **Family:** Lophopidae
- **Order:** Hemiptera.
- Jet like body posture.
- The leaf hoppers are very agile and jumps around in large numbers making a patient noise when a person walks through heavily infested field.
- Owing to the loss of cell sap the leaves turn pale yellow and shrivel up later.
- Leaf hoppers secrete a thick transparent liquid known as honey dew on which black mould is developed.
- Leaves acquire a sickly black appearance, and the attacked crop can be spoiled from a distance.

❖ **Management:**

- Spray Malathion 50 EC @ 1 liter/ha.
- **Egg parasitoid:** *Epiricania melanoleuca* (single parasitoid belonging to order Lepidoptera).

Common name	Scientific name	Family	Order
Sugar cane white fly	<i>Aleurobolus barodensis</i>	Aleyrodidae	Hemiptera
Sugar cane mealy bug	<i>Saccharicoccus sacchari</i>	Pseudococcidae	Hemiptera

Sugar cane shoot borer	<i>Chilo infuscatellus</i>	Pyralidae	Lepidoptera
Gurdaspur borer	<i>Acigona steinella</i>	Pyralidae	Lepidoptera
Sugar cane internode borer	<i>Chilo sacchariphagus indicus</i>	Pyralidae	Lepidoptera
Sugar cane scale insect	<i>Melonapsis glomerata</i>		

Insect -Pests of Pulses

Common name	Scientific name	Family
Gram pod borer	<i>Helicoverpa armigera</i>	Noctuidae
Gram cut worm	<i>Agrotis ipsilon</i>	Noctuidae
Red gram pod fly	<i>Melanagromyza obtusa</i>	Agromyzidae
Gram pod borer	<i>Helicoverpa armigera</i>	Noctuidae
Gram cut worm	<i>Agrotis ipsilon</i>	Noctuidae

1. Gram pod borer: *Helicoverpa armigera*

- **Family:** Noctuidae
- **Order:** Lepidoptera
- It is polyphagous & cosmopolitan in nature.
- Caterpillar feed foliage, when young and on seeds in the later stages.
- Larvae pupate in soil.

❖ **Management:**

- Timely sowing (up to mid-October).
- Grow early maturity varieties.
- Mixed/intercropping with barley, wheat, mustard.
- Application of NPV @ 250-500 LE/ha. (RPSC, A0-09)
- **Egg parasitoid** *Trichogramma chilonis*.

- **Larval parasitoid** of *Helicoverpa* is *Cosmopletis chloridae*.
- Use of light traps and Pheromone traps.
- Application of Monocrotophos @ 1.25 liter/ha at evening time.

2. Gram cut worm: *Agrotis ipsilon*

- **Family:** Noctuidae
- **Order:** Lepidoptera
- Polyphagous pest but cause much damage to gram crop.
- Cannibalism habit present in larvae.
- The young larvae feed on the epidermis of leaves.
- During daytime larvae live in cracks and holes in ground and come out at night and fall the plants by cutting their stem either below the surface or above the ground.

❖ **Management:**

- Hand picking & destruction of larvae.
- Application of Flubendiamide (Fume) @ 0.5 % sprays at evening time.
- Application of Malathion 5 % dust @ 25 kg/ha at evening time

3. Red gram pod fly: *Melanagromyza obtuse*

- **Family:** Agromyzidae
- **Order:** Diptera.
- The maggots eat away only a part of the seed.
- The partially damaged seeds become subjected to bacterial and fungal infection.
- Grains are rendered, unfit for human consumption.
- Maggots pupate in pods.

❖ **Management:** Application of Dimethoate @ 1.0 liter in 250-liter water/ha.

4. Other pests of Pulses:

Common name	Scientific name	Family	Order	Remarks
Tur pod bug	<i>Clavigralla gibbosa</i>	Coreidae	Hemiptera	Both nymph and adult suck the cell sap on leaves and pods of Arhar, cow pea and gram
Plume moth	<i>Exelastis atomosa</i>	Pterophoridae	Lepidoptera	It is specific pest of Arhar. Caterpillar scrapes of pods and make hole on pods.
Girdle beetle	<i>Oberia bravis</i>	Cerambycidae	Coleoptera	It is specific pest of soybean. The ovipositing female beetle girdles the stem twice. It makes three punctures just above the lower ring before inserting a single egg through the largest hole into pith. Management: spray methyl parathion 50 EC 1.25 liter/ha.
Bean fly	<i>Ophiomyia phaseoli</i>	Agromyzidae	Diptera	Feed on Phaseolus, cow pea, soybean, Arhar
Lentil pod borer	<i>Etiella zinckenella</i>	Phycitidae	Lepidoptera	

Diseases and Enemies of Honeybees

1. Predatory wasps:

<i>Vespa auraria</i>	Nests on treetops/buildings
<i>Vespa magnifica</i>	Under-ground nest
<i>Vespa cincta</i>	Underground nest
<i>Vespa basalis</i>	Nest on treetop/buildings

- The wasps catch the bees at hive entrance and kill them.
- Most serious damage in hills is caused by *V. magnifica* which cuts down bees in large number.
- The weak colonies may even perish due to its attack.
- Burn the nests during nighttime.
- Kill the wasps in the apiary by flapping

❖ **Wax moth (*Galleria mellonella*)**

❖ **Nature and extent of damage:**

- The attack is more prevalent during monsoon.
- The wax moth larvae tunnel through the mid ribs of the comb and there is presence of small mass of minute wax particles outside the tunnels.
- In case of severe infestation, further brood rearing is stopped; bees stop field work and colony may abscond.

❖ **Prevention and control:**

- Close cracks and crevices in the hive. Reduce hive entrance.
- Remove combs not covered by bees. Keep the bottom board clean.

❖ **Control in storage**

- Keep spare combs in empty hive bodies in tiers and close both at bottom and top
- Disinfect the stack by burning sulphur @ 180 g/ cubic metre (fumigation by sulphur fumes).
- After fumigation, put naphthalene flakes in moth proof stacks

❖ **Ectoparasitic mites**

- In India, ectoparasitic mites *Varroa destructor* and *Tropilaelaps clareae* are causing severe damage to *A. mellifera* colonies.
- No damage in *A. cerana* colonies due to these mites.

Diseases in honeybees caused by

❖ **Spread of diseases**

- Most of the diseases spread among bees and from colony to colony or apiary to apiary through: Robbing, drifting, absconding, manipulation and migration; hence need proper management

❖ **Bacterial diseases**

❖ **European foul brood**

- First isolated report in 1971 in *Apis cerana*
- Late 1990's: in *A. mellifera* from North India; now widespread
- **Causative organism:** *Melissococcus pluton*
- Infected larvae in coiled stage; displaced in cells; die when 4-5 day old; dead brood mostly in uncapped stage.
- Colour of brood changes from shiny white to pale yellow then brown; on drying form rubbery scales.
- Sour Odour from dead brood; no ropiness in decaying larvae
- Worker, drone and queen; all type of brood affected
- **Control:** Feed Terramycin @200mg in 500 ml conc. sugar syrup to affected colony

❖ American foul brood

- **Causative organism:** *Paenibacillus larvae*
- **Time of death:** Late larval or early pupal stage
- **Position of dead brood:** Lying flat on cell base
- **Type of brood:** Worker, rarely drone or queen
- **Control:** Terramycin 0.25 – 0.4g in 5lt sugar syrup feeding

❖ Protozoa and mite diseases of adult bees

❖ Nosema; Protozoan disease

- **Nosema:** Severe during spring and winter
- Bees become dysentric, droppings of yellow excreta on combs or on parts of hive
- Infects all castes of bees; queen stops egg laying
- *Nosema apis* (protozoan)
- infected bees collect in front of hive, sluggish, crawlers on leaf blades, distended abdomen
- Feed fumigillin 200 mg in sugar syrup to each colony or two feedings at weekly interval of Dependel-M @ 0.5g/ litre/ colony

❖ Diseases of Mulberry Silkworm

1. Viral Disease

➤ Symptom:

- ✓ The larvae will be sluggish with swollen intersegmental region.
- ✓ The integument of diseased larvae will be fragile and breaks easily.
- ✓ On injury milky fluid containing many polyhedral inclusion bodies oozes out from the larval body.
- ✓ The diseased larvae do not settle for moult and show shining integument.
- ✓ The larvae appear to be restless.
- ✓ The dead larvae hang by hind legs heads downward

➤ Management

- ✓ Sun drying of rearing appliances for one/two days.
- ✓ Disinfection of rearing room and appliances with 5% bleaching powder.
- ✓ Disinfection of worms, trays and discarding of diseased worms
- ✓ Ensure proper ventilation and air circulation.
- ✓ Provide proper bed spacing.
- ✓ Feed the larvae with nutritious mulberry leaves.
- ✓ Collect and burn infected larvae, faecal matter and bed refuses.
- ✓ Early diagnosis and rejection of infected lots.
- ✓ Dust the bed disinfectant, Vijetha (or) Resham Keet Oushadh on the larvae, after each moult and ½ hr. before resumption of feeding (3 kg/100 dfl).
- ✓ Spray 1% of extract of *Psoralea coryleifolia* on mulberry leaves, shade dry and feed worms once during third instars.

2. Bacterial Diseases

- Bacteria and viruses cause the disease individually or in combination.
- Fluctuating temperature and humidity and poor-quality mulberry predispose the disease development.
- The diseased larvae will be stunted in growth, will be lethargic soft and appear flaccid.
- The cephalothoracic region may be translucent.
- The larvae vomit gut juice develop dysentery and excrete chain type feces.

- The larvae on death putrefy, develop different and emit foul smell

- **Management**

- ✓ Maintenance of hygienic condition.
- ✓ Disinfection of rearing room and appliance.
- ✓ Disinfection of worms, trace and discarding of sick wormsAvoid injury to the worms, overcrowding of trays and accumulation of faeces in the rearing bed.
- ✓ Sound management, improving the rearing environment and feed stuff.
- ✓ Feeding the larvae with healthy nutritious leaves.
- ✓ Early diagnosis and rejection of infected lots.
- ✓ Avoid spraying commercial **Bt** insecticides in nearby mulberry field.
- ✓ Apply antibiotics like Streptomycin/Tetracycline/Ampicillin

i. White muscadine

- It is caused by a fungus *Beauveria bassiana* and the green muscadine is caused by a fungus *Spicaria prasina*.
- Aspergillosis is common in young age silkworms and the infected larvae will be lustrous and die.
- Dark green (*Aspergillus flavus*) or rusty brown (*Aspergillus tamari*) mycelial cluster are seen on the dead body.
- The diseases larvae prior to death will be lethargic and on death are flaccid oil specks may be seen on the surface of larvae.
- They gradually be foam hard, dry, and mummify into a white or green colored structure.
- The diseases pupae will be hard, lighter and mummifies.
- **Management:**
 - ✓ Sundry the rearing appliances.
 - ✓ Disinfect the rearing room and utensils with 5 per cent bleaching powder.
 - ✓ Avoid low temperature and high humidity in the rearing room.
 - ✓ Keep the rearing bed thin and dry.
 - ✓ Early diagnosis and rejection of infected lots.
 - ✓ Apply Dithane M45 (3 kg/100 dfls) / Vijetha supplement as disinfectant on the larvae.
 - ✓ Disinfect rearing rooms and trays with 4 per cent pentachlorophenol to control Aspergillosis.

3. Protozoan disease:

i. Pebrine

- Diseases larvae show slow growth, undersized body and poor appetite.
- Diseases larvae reveal pale and flaccid body. Tiny black spots appear on larval integument.
- Dead larvae remain rubbery and do not undergo putrefaction shortly after death.
- **Management of Pebrine:**
 - ✓ Produce healthy eggs.
 - ✓ Disinfection of rearing room and utensils.
 - ✓ Maintain strict hygienic conditions during rearing
 - ✓ Surface disinfect the laying in 2 per cent formalin for 10 minutes before incubation.
 - ✓ Collect and burn the diseased eggs, larvae, pupae and moths, bed refuses, faecal pellets, etc.

1. Mango hoppers

- *Amritodus atkinsoni* (**largest**),
- *Idioscopus clypealis* (**smallest**),
- *Idioscopus niveosparsus* (**medium**)
- **Family:** Cicadellidae
- **Order:** Hemiptera
- **Host:** Mango
- **These insects are monophagous on mango and occur as regular pests.**

❖ Symptoms and Nature of Damage:

- During flowering, the hoppers develop enormously in number, suck juice from the inflorescence and other tender plant parts reducing the vigour of the plant leading to reduction in fruit set and even premature fruit fall.
- The infestation also leads to development of sooty mould on the honeydew excreted by the insects.
- Egg laying also inflicts injury to the inflorescence. The infestation ranges from 25 to 50 per cent and in severe case it may lead to total loss of crop.
- During the remaining part of the year (off season), these hoppers occur in small numbers inside barks or on leaves of mango. High humidity in the air during flowering time encourages insect multiplication.
- Affected flowers wither and drop down.
- Presence of black sooty mould on floral and other tender plant parts.
- High humidity in orchards due to waterlogging, shading and overcrowding of trees favour the build-up of the pest.
- There are two peak generations of this insect during a year i.e. during February – April and June-August periods.

❖ Management

- Keeping orchard clean.
- Avoiding overcrowding and waterlogging.

- Proper pruning of the tree after harvesting to facilitate proper sunlight and air that minimises hopper population

❖ **Spray schedule is recommended as follows**

- At flower bud initiation, imidacloprid (0.036%) or carbaryl (0.1%)
- At emergence of inflorescence stalks and before flower opening (anthesis), dimethoate (0.03%) or thiamethoxam (0.025%) or imidacloprid (0.0036%).
- During anthesis and pollination, insecticides should not be sprayed.
- In case of very high population of hoppers at this stage, endosulfan 2ml/l is recommended
- When fruits are of pea size, monocrotophos (0.036WSC) or dimethoate (0.03%)
- Adding sulphur 3.5g/l to the insecticide based on need to check mites and sooty mould, *Capnodium mangiferum*
- Directing the spray first to stem/ trunk, then branches, twigs, leaves and finally inflorescence is a recommended method.

2. **Mango stem borer: *Batocera rufomaculata***

- **Family:** Cerambycidae
- **Order:** Coleoptera
- It is a **polyphagous** pest, infesting besides mango, apple, fig, mulberry, Eucalyptus, jack fruit, papaya etc.

❖ **Symptoms of Damage**

- Masses of grass and sap exuding from the bore holes.
- Leaves of damaged branches dry and fall.
- Branches collapse, tree succumbs in severe cases

❖ **Management**

- The affected portions with grubs and pupae removed and destroyed, if branches are affected.
- The bore holes traced and opened.
- A swab of cotton wool soaked in chloroform or petrol 5ml or carbon disulphide or kerosene oil or dichlorvos 10ml/l inserted into the hole and sealed with mud.
- Methyl parathion 1 ml/l poured into the hole or tablet of aluminium phosphide inserted into the hole to kill the grub.

- When burrows are superficial, extract the grubs with stiff hooked wire and paint bordeaux paste.

3. Mango fruit fly *Bactrocera dorsalis*

- **Family:** Tephritidae
- **Order:** Diptera
- It is one of the major pests of mango in India. It also infests guava, peach, citrus, ber, banana, papaya etc.

❖ Symptoms & Damage

- Semi ripe fruits with decayed spots.
- Dropping of fruits Damage to semi ripe fruits is caused by both maggot and the adult.
- The oviposition punctures made by the female serves as entry for fermenting organisms.
- Maggots feed on the pulp and convert the pulp into bad smelling discoloured semi liquid mass, unfit for use.
- The fruits develop brown rotten patches on them and fall to the ground eventually.

❖ Management

- Collection and destruction of fallen rotten fruits.
- Raking under the trees to expose the pupae.
- Mixing of carbaryl 10D in soil @ 50-100 g/tree.
- Install pheromone traps.
- Foliar spray with malathion (0.1%) + gur (2%) a month before harvesting the fruit crop repeated after 15 days.

❖ Post-Harvest Control (Heat treatment techniques):

- **Hot water treatment:** Submerging fruits in hot water at 43 to 46.7°C for 35- 90 min.
- **Double dip method:** Immersion of mango fruits in water at 40°C for 20 minutes, followed by 10 minutes at 46°C to get 100 per cent mortality of *Bactrocera dorsalis* eggs.

4. Mango mealybug: *Drosicha mangiferae*

- **Family:** Pseudococcidae
- **Order:** Hemiptera
 - ❖ **Symptoms & Damage**
- Both nymph and adults suck sap from other tender plant parts thus reducing the plant vigour.
- Female lays eggs in clusters within ovisacs in soil under the trees 5-15 cm deep during April and May.
- The nymphs ascend the trees and settle on inflorescence.
- Nymphs and adults suck sap from inflorescence, fruit stalks, fruits *etc.* leading to flower drop, premature fruit drop *etc.*
- They also excrete honey dew on which sooty mould develops and the fruit development is hampered.

❖ **Management**

- Deep summer ploughing up to base of the tree trunks, after harvesting to expose eggs of mealy bugs.
- Dusting methyl parathion 2D or chlorpyrifos 5D around tree and incorporating into the soil.
- Spraying with acetamiprid (0.05%) or imidacloprid (0.0036%) or monocrotophos (0.036%), when severe mealybug infestation noticed on the twigs.
- Wrapping 25 cm wide, 400-gauge polythene sheet on the tree trunk 30 cm above ground level and pasting grease over it to prevent migration of freshly hatched first instar nymphs during winter (Nov-Dec) from soil to trees, one week before their emergence.
- Crawlers collecting beneath the polythene sheet may be scraped with a knife.

5. Other Pests

Common name	Scientific name	Family	Order
Mango Psylla	<i>Asylla cistellata</i>		
Mango Gall Midge	<i>Amradiplosis echinogalliperda</i> ,	Cecidomyiidae	Diptera

	<i>Procontarinia matteiana</i>		
Mango nut weevil or stone weevil	<i>Sternochetus mangiferae</i>	Curculionidae	Coleoptera
Mango fruit borer	<i>Deanolis albizonalis</i>	Pyralidae	Lepidoptera
Aphid	<i>Toxoptera odinae</i>	Aphididae	Hemiptera
Mango leaf webber	<i>Orthaga exvinacea</i>	Noctuidae	Lepidoptera
Termites	<i>Odontotermes obesus</i>	Termitidae	Isoptera

6. Pomegranate butterfly / Anar butterfly *Deudorix isocrates*

- **Family:** Lycaenidae
- **Order:** Lepidoptera
- It is the most important and destructive pest of pomegranate and distributed throughout the country, also infesting guava, annona, apple, ber, citrus, litchi, tamarind, wood apple, soap nut, etc.

❖ Symptoms & Damage

- The conspicuous symptoms are offensive smell and excreta of caterpillar at the entry hole.
- The affected fruits ultimately falling down.
- The fruit appears healthy but the caterpillar inside feeds on pulp and seeds just below the rind.
- It is only when the grown-up caterpillar comes out, a round hole is seen through which juices come out.

❖ Management

- Destruction of fallen infested fruits checks the spread.
- Removal of flowering weeds especially of Compositae family.
- Though expensive, bagging of fruits with polythene or paper bags or cloth bags soon after the fruit set prevents the pest attack.
- Initiate the spray schedule with the onset of flowering with any of following insecticides:
- Deltamethrin (0.0028%), cypermethrin (0.0075%), monocrotophos (0.036%), carbaryl (0.1%).

- Repeat the spray at 15-20-day interval.
- About 3 to 4 sprays are needed for effective control of the pest, as it continues to attack flowers (flowering in pomegranate remains for a longer time).

7. Guava mealy bug *Ferrisia virgata*

- **Family:** Pseudococcidae
- **Order:** Hemiptera
- It is otherwise called white tailed mealy bug /striped mealy bug.
- It is a widely distributed species in tropical and subtropical countries.
- It also infests bhindi, amaranthus, coccinia, colocasia, lab lab, tomato, brinjal, cashew, anona, guava, amla, grape, tobacco, pepper, cotton, **betelvine**

Pests of Grapevine

1. Grapevine Flea Beetle *Scelodonta strigicollis*

- **Family:** Chrysomelidae
- **Order:** Coleoptera
- It is the most destructive pest of grapevine all over India.

❖ Symptoms & Damage

- They feed on mature leaves cutting elongated holes on the leaf lamina like shot holes.
- The damage results in: Complete fed sprouting buds, Shot holes (rectangular cuttings) on mature leaves.
- Adult beetles hibernate during winter under tree bark and become active from March till November.
- Adults have characteristic habit of falling down and feigning death when disturbed.

❖ Management

- Adult beetles may be collected and killed.
- Removal of loose bark in rainy season after pruning to expose and eliminate eggs and adults found underneath.

- First spraying when buds swell in early morning or evening hours to kill beetles and second spray after 10 days with monocrotophos (0.036%) or carbaryl (0.1%) or imidachloprid (0.0072%) or cypermethrin (0.0075%)

2. Grapevine thrips: *Rhipiphorothrips cruentatus*

- **Family:** Thripidae
- **Order:** Thysanoptera
- Most destructive pest of grapevine India. It also feeds on rose, jasmine, cashew and other fruit trees.

❖ Symptoms & Damage

- Young nymphs on hatching feed on the under surface of leaves.
- Both the nymphs and adults' lacerate tender foliage and suck the oozing sap.
- The attacked leaves appear silvery initially and later turn brown and give withered appearance, curl up and drop off the plants.
- Severely affected vines do not bear fruits. If fruits are attacked, they develop corky layer on the fruits and turn brown.
- Infestation results in: Silvery patches on the affected leaves, Brown corky patches on fruits (scab)

❖ Management

- Removal of weeds in and around garden.
- Cutting of infested branches and burning.
- Spraying dimethoate (0.03%) or methyl demeton (0.025%) or thiamethoxam (0.25%)

3. Citrus butterfly *Papilio demoleus*, *Papilio polytes*, *Papilio helenus*

- **Family:** Papilionidae
- **Order:** Lepidoptera
- **Host:** It infests almost all citrus varieties though Malta (*Citrus sinensis*) is its preferred host.

- It can feed and breed on all varieties of cultivated or wild citrus and various other species of family Rutaceae.
- Besides citrus, it also attacks ber, wood apple, curry leaf.

❖ Symptoms and Damage

- Leaves fed up to midribs.
- Freshly hatched caterpillars are dark brown and soon develop irregular white markings on their body resembling bird's drop.
- The caterpillars feed voraciously on tender leaves right up to the mid ribs and defoliate the entire seedlings or the tree leaving behind the only midribs.

❖ Management

- In small orchards and nurseries with mild infestation, hand picking and destruction of various stages of the pest.
- Natural enemies enumerated below suppress the pest population
- **Egg parasitoids:** *Trichogramma evanescens*; *Telenomus* sp.
- **Larval parasitoid:** *Distatrix papilionis*; *Brachymeria* sp.
- **Pupal parasitoid:** *Pterolus* sp.
- Spraying of monocrotophos (0.036%) or cypermethrin (0.0075%) or deltamethrin (0.0028%)
- When the caterpillars are small. *B. t.* formulation HALT at 9 g/l is also recommended

4. Citrus leaf miner *Phyllocnistis citrella*

- **Family:** Gracillariidae
- **Order:** Lepidoptera
- It attacks all species of citrus but prefers sweet oranges. It also infests jasmine etc.

❖ Symptoms & Damage

- Characteristic silvery white zigzag galleries below the epidermis of tender leaves.
- Serious infestation causes retardation in growth. The infestation predisposes the leaves to canker growth.

- The tiny red, apodous larva mines in between the epidermal layers of the leaf in a zigzag manner and feeds on chlorophyll which results in distortion of the leaf lamina.
- The tunnel appears silvery white. New and tender leaves are preferred.
- Sometimes, the larva mines the outer layer of young green twigs.
- The full-grown caterpillar is slender, yellowish green with brownish mandibles.
- It is found inside the gallery formed in leaf tissue. Larval period is 15-30 days.
- Pupation takes place inside the leaf mine. Pupal period is 5-25 days.
- Total life cycle takes on an average 20-60 days depending on the climate.
- There are 9-13 overlapping generations in a year.

❖ **Management**

- Pruning of affected parts during winter and burning.
- Spraying of Neem cake solution 5% or neem oil 5% or monocrotophos (0.036%) or dimethoate (0.030%) or imidachloprid (0.0036%) twice at 10 days interval at every new flush time i.e., during June – July, Sep – Oct, Dec – Jan.

Common name	Scientific name	Family	Order
Citrus fruit sucking moths	Eudocima materna, Eudocima fullonica, Eudocima ancilla	Noctuidae	Lepidoptera
Citrus rust mite	Phyllocoptruta oleivora	Eriophyidae	Acarina
Citrus blackfly	Aleurocanthus wogluni	Aleurodidae	Hemiptera

PAPAYA

❖ **White fly, *Bemisia tabaci***

❖ **Symptoms of damage**

- Nymphs and adults suck the sap from under surface of the leaves
- Yellowing of leaves.

❖ **Management**

- Field sanitation

- Removal of host plants
- Installation of yellow sticky traps
- Spray application of imidacloprid (0.0036%) or triazophos (0.04%) during heavy infestation.
- Spray neem oil 3% or NSKE 5%
- Release of predators viz., Coccinellid predator, *Cryptolaemus montrouzieri*
- Release of parasitoids viz., *Encarsia haitierrsis* and *E.guadeloupae*

INSECT-PESTS OF TEMPERATE FRUITS

Sanjose scale	Apple shot hole borer	Hairy caterpillars	Apple leaf blotch miner
Woolly apple aphid	Apple stem borer	Thrips	Apple leaf hoppers
Apple root borer	Apple fruit moth	Apple fruit rhynhites	Apricot fruit moth

1. Sanjose scale *Quadraspidiotus perniciosus*

- **Order:** Homoptera
- **Family:** Diaspidae
- **Host:** 200 species, Rosaceae family (apple, pear, peach, plum etc.)
- ❖ **Symptoms:**
 - Less infested trees show small, grey specks on the bark surface.
 - Severely infested trees have the bark covered with grey layer of over-lapping scales appearing as if sprayed wood ash.
 - Red inflamed areas on fruits.
- ❖ **Nature of Damage**
 - **Suck** cell sap, as a result , the young plants in the nursery become weak and ultimately die.
 - Stems and branches of trees have poor growth, low vigor and ultimately death owing to sucking sap

- Infested fruits have more scales at the stem end and the calyx
- Marketing value of the fruits is greatly hampered due to infestation

❖ **Management**

- Spray between 1/2" green stage and tight cluster stage with TSO
- Last week of May-First week of June spray 0.04% chlorpyrifos/0.025% Metasystox/Summer oil
- Post-harvest spray oil 0.04% chlorpyrifos

2. Woolly apple aphid: *Eriosoma lanigerum*

- **Order: Homoptera**
- **Family: Aphididae**
- **Host: Apple, crab apple**
- Aerial parts-white woolly mass
- Roots-gall formation

❖ **Nature of Damage**

- Sucking sp from stems, twigs and roots.
- Affected plants-stunted growth, loss of vigor, reducing fruiting and gall formation
- Severe galls in the roots cause small size fruits and early coloration
- Pest over winters as nymphs in crevices, loose bark and underground parts
- Reproduction parthenogenetically

❖ **Management**

- Non-Bearing Trees: Phorate 10-30g or carbofuran 30-50g at 5cm depth of soil
- Bearing Trees: Spray in June-October 0.04% chlorpyrifos or 0.08% triazophos
- Drench same during April and Sep/Oct
- Apply 25-30g phorate or 70-80g carbofuran
- *Aphelinus mali* Biocontrol agent
- Remove water shoots
- M-25 rootstock resistant
- Malling Merton series resistant (Kotkhai research)

COCONUT

1. Rhinoceros beetle: *Oryctes rhinoceros*

❖ **Symptoms of damage**

- Central spindle appears cut or toppled

- Fully opened fronds showing characteristic diamond shaped cuttings
- Holes with chewed fibre sticking out at the base of central spindle.

❖ **Management**

- Remove and burn all dead coconut trees in the garden (which are likely to serve as breeding ground) to maintain good sanitation.
- Incorporate the entomopathogen i.e., fungus (*Metarrhizium anisopliae*) in manure pits to check the perpetuation of the pest.
- Soak castor cake at 1 kg in 5 l of water in small mud pots and keep them in the coconut gardens to attract and kill the adults.
- For seedlings, apply 3 naphthalene balls/palm weighing 3.5 g each at the base of inter space in leaf sheath in the 3 inner most leaves of the crown once in 45 days.
- Set up light traps following the first rains in summer and monsoon period to attract and kill the adult beetles.
- Field release of Baculovirus inoculated adult rhinoceros beetle @ 15/ha reduces the leaf and crown damage caused by this beetle.
- Apply mixture of either neem seed powder + sand (1:2) @150 g per palm or neem seed kernel powder + sand (1:2) @150 g per palm in the base of the 3 inner most leaves in the crown.
- Place phorate 10 G 5 g in perforated sachets in two inner most leaf axils for 2 times at 6 months intervals.
- Set up rhinolure pheromone trap @ 1 per ha to trap and kill the beetles.
- Maintain good sanitation soak castor cake at 1 kg in 5 litres of water in small mud pots and keep them in coconut gardens to attack and kill adults.
- Apply 3 Naphthalene balls /palm weighing 3.5g each at the base of the inter space.

2. Red palm weevil: *Rhynchophorus ferrugineus*

❖ **Symptoms of damage**

- Holes on trunk with brownish ooze.

- Yellowing of inner leaves.
- Gradual wilting of central shoot in the crown.
- Remove and burn all wilting or damaged palms in coconut gardens to prevent further perpetuation of the pest.
- Avoid injuries on stems of palms as the wounds may serve as oviposition sites for the weevil. Fill all holes in the stem with cement.
- Avoid the cutting of green leaves. If needed, they should be cut about 120 cm away from the stem.
- Fill the crown and the axils of topmost three leaves with a mixture of fine sand and neem seed powder or neem seed kernel powder (2:1) or lindane 1.3 D (1:1 by volume) once in three months to prevent the attack of rhinoceros beetle damage in which the red palm weevil lays eggs.
- Plug all holes and inject pyrocone E or carbaryl 1% or 10 ml of monocrotophos into the stem by drilling a hole above the points of attack.
- Setting up of attractant traps (mud pots) containing sugarcane molasses 2½ kg or toddy 2½ litres + acetic acid 5 ml + yeast 5 g + longitudinally split tender coconut stem/logs of green petiole of leaves of 30 numbers in one acre to trap adult red palm weevils in large numbers.
- Install pheromone trap @ 12 per ha

3. Black headed caterpillar: *Opisina arenosella*

❖ Symptoms of Damage

- Dried up patches on leaflets of the lower leaves
- Galleries of silk and frass on underside of leaflets

❖ Management

- Among the larval parasitoids, the bethylid *Goniozus nephantidis* is the most effective in controlling the pest.
- The optimum level of release is 1:8 of host-parasitoid ratio.

- The parasitoid should be released @3000/ha under the coconut trees when the pest is in the 2nd or 3rd instar larval stage. Parasitoid release trap may be used to release the parasitoid at the site of feeding.
- Parasitoids should not be released in the crown region since they will be killed by predators like spiders and reduviid bugs.
- Remove and burn all affected leaves/leaflets.
- Spray malathion 50 EC 0.05% (1ml/lit) to cover the undersurface of the leaves thoroughly in case of severe epidemic outbreak of the pest in young palms.

TEA

1. Tea mosquitobug, *Helopeltis theivora*

- It is considered to be a serious pest of a tea.
- Adults and nymphs suck the sap from buds, young leaves and tender stems.
- Puncturing the plant tissues with their needle like rostrum and inject toxic saliva.
- Punctures appear as reddish-brown spots.
- Leaves curl up, badly deformed and shoots dry up.

2. Tea Mites

Common Name	Scientific Name
Red spider mite	Oligonychus coffeae
Scarlet mite	Brevipalpus californicus
Purple mite	Calacarus carinatus
Pink mite (or) Orange mite	Acaphylla theae
Yellow mite	Polyphagotarsonemus latus

❖ Management

- Spray dicofol 18EC 2 ml/lit or ethion 50 EC 2 ml/lit or monocrotophos 1ml/lit
- Application of wettable sulphur 80 WP 2g/lit using hand operated sprayer.

1. Coffee berry borer, *Hypothenemus hampei*

- In India, the pest was first noticed on a few plantations in the Nilgiris during early 1990.
- Pin hole at the tip of the berries (navel region)
- Severe infestation - two or more holes may be seen.
- Female beetle bores into the berries through the navel region make tunneling and feed inside content.
- Powdery substance pushed out through the holes

❖ Management

- Proper adoption of cultural practices and phytosanitary measures important for management of coffee berry borer.
- Transportation of infested coffee to uninfested areas is the main reason for spread.
- Crop bags should be fumigated before delivery to estates to avoid cross infestation.
- Timely harvest
- Gleaning - spreading gunny bags or polythene sheets on the ground after picking the berries - minimize gleaning.
- Maintain optimum shade and good drainage.
- Dipping infested berries in boiling water for 2-3 minutes kills all the stages inside.
- Drying of coffee beans – prevents breeding of beetles in stored coffee based on moisture content level.
- Arabica (10% moisture content) - 16kg/lit
- Robusta (11% moisture content) - 18kg/lit
- Install attractant trap to collect and kill the beetles

SOME IMPORTANT INSECTS OF MAJOR CROPS

Rice		
Stem borers	Yellow stem borer	Scirpophaga incertulas
	Pale headed borer	Chilo suppressalis
	Dark headed striped borer	Chilo polychrysa
	Pink stem borer	Sesamia inferens
	Sugarcane shoot borer	Chilo infuscatellus
Leaf feeders	Rice gall midge	Orseolia oryzae
	Rice caseworm	Nymphula depunctalis
	Rice ear cutting caterpillar	Mythimna separata
	Rice hispa	Diclodispa armigera
	rice grasshopper	Hieroglyphus banian
Sap suckers	rice bug	Leptocoriosa acuta

	Brown planthopper	Nilaparvata lugens
	White backed plant hopper	Sogatella furcifera
Important Pests of Vegetable crops		
Brinjal: Brinjal shoot and fruit borer	Lucinodes orbonalis	
Potato: Potato Tuber Moth	Phthorimaea operculella	
Sweet potato: Sweet potato weevil	Cylas formicarius	
Tomato: Tomato fruit borer	Helicoverpa armigera	
Okra: Spotted bollworms	Earias insulana, E. vitelli	
Red cotton bug	Dysdercus koenigii	
Cabbage and Cauliflower		
Diamond Back Moth	Plutella Xylostella	
Cabbage butterfly	Pieris brassicae	
Gourds		
Red pumpkin beetle	Aulacophora foveicollis	
Melon fruit fly	Bacterocera cucurbitae	
Onion		
Onion thrips	Thrips tabaci	
Onion maggot	Delia antiqua	

❖ **Polyphagous pests:** Locust, grasshopper, termites, hairy caterpillars, cutworms, thrips

1. Locust: Patanga succincta (Bombay locust)

- Locust migratoria (Migratory locust)
- Schistocera gregaria (Desert locust)

2. Grasshopper

- Oxya nitidula (Sporadic occurrence)

DEFENCE MECHANISMS OF INSECTS

I. Behavioural:

- (i) **Thanatosis** - insects pretends as if dead. e.g., some beetles
- (ii) **Structural** e.g., hardened forewings of beetles known as elytra protect the beetles from predation of birds.
- (iii) **Colourational:** Presence of protective colours. e.g., Stick insects
- (iv) **Chemical:** Presence of defensive chemicals. e.g., Bees producing venom

❖ **The following are some examples of waiting period of some chemicals in a few important crops:**

Crop	Insecticide	Dose	Waiting period (days)
Chillies	Dicofol	0.05%	1
	Quinalphos	0.05%	8
Tomato	Phosalone	0.05%	3
	Quinalphos	0.05%	5
Brinjal	Phosalone	0.05%	2
	Endosulfan	0.07%	3
	Aldicarb	1 kg a.i./ha	60

FACTS FROM PREVIOUS YEAR EXAMS

- ❖ At which stage of Gram crop is damaged by *Helicoverpa armigera*: **Pod formation stage (AFO-2015)**
- ❖ Barium carbonate and aluminum phosphate are generally used in: **Rodenticide (AFO-2016)**
- ❖ Khapra beetle is pest of: **Stored grain (AFO-2016)**
- ❖ Trichogramma is: **An insect (AFO-2016)**
- ❖ Aluminum phosphide is a kind of: **Rodenticide (AFO-2017)**
- ❖ Khapra beetle is related to: **Storage food grain pest (AFO-2017)**
- ❖ Which among the following insect causes damage inside the stem: **Stem borer (AFO-2019)**
- ❖ Aphid is responsible for which disease in tobacco: **Cucumber mosaic virus (AFO-2020)**
- ❖ About Honeybee and bee colony, which of the following is not correct: **Unfertilized egg become worker bee (AFO-2020)**
- ❖ Guava pest which deposit eggs on soft skin of ripening fruit, on hatching maggot bore into fruit and feed on the soft pulp, the infested show depressions with dark green puncture and when cut opens the maggots are visible and finally fruit rot and fall: **Fruit fly (AFO-2020)**
- ❖ Sorghum shoots fly attack symptoms: **The larva cuts the growing point, resulting in wilting and drying of the central leaf (AFO-2020)**
- ❖ Which trap is used to trap male in IPM: **Pheromone trap (AFO-2020)**
- ❖ Which entomopathogens fungus role as biological control agent in cotton bollworm: **Beauveria bassiana (AFO-2020)**
- ❖ Which bee is best for domestication. This bee is also known as Italian bee: **Apis mellifera (AFO-2021)**
- ❖ Piercing and sucking type mouth part in which order: **Hemiptera (AFO-2021)**
- ❖ Beetle having horn like projection on head, black body and attacks on open terminal portion: **Rhinoceros beetle (AFO-2021)**
- ❖ Leaf curl of chilli is transmitted by: **White fly (AFO-2022)**
- ❖ What are the diagnostic symptom of infestation of stem borer of maize: **Caterpillar feed on the leaves and bore into stem and cob (AFO-2022)**
- ❖ 1. What is the bio agent to control common mealy bug: **Leptomastix dectylopii (RRB SO-2019)**
- ❖ Which disease of groundnut is caused due to thrips: **Bud necrosis (RRB SO-2019)**
- ❖ Which among the following is the most serious pest of cotton which eat cotton boll and boll will fall down after attack and remaining unable to open properly and moth will be brown in colour, and larvae will be pink in colour which puncture into the boll: **Pink bollworm (RRB SO-2020)**
- ❖ Q1. Thrips transmitting plant virus belongs to: **Tospo virus (RRB SO-2021)**